

Understanding linguistic evolution using an event-based model

Anonymous submission

Abstract

Linguistic phylogenies are standardly inferred from abstract cognate relationships among lexemes. Despite the prevalence of this practice, it suffers from well-known drawbacks: it disregards the phylogenetic signal in the segmental form of words; the types of questions that can be addressed are limited; and it treats cognate judgments, which are hypotheses, as observations. In this study, we present a framework for comparative linguistic analysis that overcomes these deficiencies. At the heart of our framework is the TKF91 model, which is a model of molecular evolution originally developed to account for insertion and deletion events in DNA sequences. Within this framework phylogenetic inferences can be drawn directly from cognate word forms, which opens a new horizon of possibilities for historical and evolutionary linguistics. We showcase in particular the ability of this framework to deliver new insights into sound change and the evolution of segmental inventories.

Key words: Bayesian inference, Language Evolution, Phylogenetics, Markov chain Monte Carlo, Phonology, Romance

1. Introduction

Natural languages exhibit the property of descent with modification, which allows their histories to be represented by a tree structure and estimated using phylogenetic methods developed in evolutionary biology (Atkinson et al. 2005; Croft 2008; Pagel 2009; Borchsenius et al. 2017; Bromham 2017; Pagel 2017). Over the past twenty years, these methods have come to play a prominent role in historical and evolutionary linguistics. Bayesian phylogenetic methods in particular are increasingly used to infer tree topologies, ancestral states, and divergence times (Bouckaert et al. 2012; Bowerman et al. 2012; Chang et al. 2015; Kim et al. 2017; Sagart et al. 2019; Carling et al. 2021; Auderset et al. 2023; Heggarty et al. 2023). Despite the undeniable advances of these methods, a fundamental problem remains unaddressed: the nature of the observations.

1.1. Root-meaning traits

Linguistic phylogenies are typically inferred from abstract cognate relationships (Gray et al. 2000;

Holden 2002; Ringe et al. 2002; Gray et al. 2003; Bouckaert et al. 2012; Bowerman et al. 2012; Chang et al. 2015). Cognates are linguistic units at different levels (e.g., segments, morphemes, words) that share a common ancestor. An example from the lexical domain is Spanish *mano* ‘hand,’ French *main*, and Italian *mano*, which all descend from Proto-Romance */manu/. Four types of lexical cognacy have been identified in the literature (Chang et al. 2015), but studies of linguistic phylogenetics rely overwhelmingly on one: root-meaning traits

Root-meaning traits encode the ancestral relationships of the root of the most semantically general and stylistically neutral word for a given concept, such as ‘hand’ (Heggarty et al. 2023). The concepts themselves are drawn from a specific subset of the lexicon commonly referred to as the “basic vocabulary,” which typically includes words for body parts, basic actions, natural elements, and other core aspects of daily life that are allegedly less likely to undergo borrowing or replacement. Swadesh lists are compilations of basic vocabulary terms, which now exist in several versions (Tadmor et al. 2010).

Table 1 presents a root-meaning trait in a sample of Indo-European languages for the concept ‘hand.’ Lexical items that share a common ancestor are assigned to the same cognate class. These assignments are based on regular sound correspondences among the languages. The multi-state cognate-class representations in Table 1 are usually transformed into binary characters, in which 0 and 1 values denote the absence and presence, respectively, of lexemes belonging to a particular cognate class. These values are treated as the observations for phylogenetic inference.

Despite their prevalence, root-meaning traits suffer from several drawbacks. First and foremost, the transformation of word forms into abstract cognate relationships results in pervasive information loss. For instance, the lexemes for ‘hand’ in Spanish, French, and Italian are all assigned the same state in Table 1. Since there is no variation, no inferences can be made about historical relationships of the languages with these lexemes. There is, however, phylogenetic signal in the forms of the words themselves, as French /mɛ̃/, Spanish /mano/, and Italian /mano/ have diverged from Latin /manum/. The current practice of linguistic phylogenetics is unable to take such differences into account.

47 Second, since root-meaning traits are analyzed the same way biologists analyze morphological characters (Lewis 2001), they are subject to the same limitations: (i) The state labels

Table 1. A root-meaning trait for the concept ‘hand.’

Language	Phonemic Representation	Cognate Class
English	/hænd/	0
German	/hant/	0
Dutch	/fiand/	0
French	/mɛ̃/	1
Spanish	/mano/	1
Italian	/mano/	1
Russian	/ruka/	2
Polish	/rɛŋka/	2
Lithuanian	/ranka/	2

(whether binary or multistate) are arbitrary, with no intrinsic meaning except to distinguish cognate classes. As a result, studies of linguistic phylogenies are restricted to models with symmetrical rates of change, in order that the probability of the observations remain the same regardless of the state label assignment; the probability of the data should remain the same if the state labels are swapped, with the 0s coded as 1s and the 1s coded as 0s. (ii) Similarly, the state labels from one word to another have different meanings. State 0 of one concept is not equivalent to state 0 of another. State information across cognates can accordingly not be pooled to learn about their evolutionary dynamics. (iii) Since invariant root-meaning traits (with a single state assigned to each language under investigation) are uncommon, probability calculations must condition on constant patterns being absent from the data matrix. Phylogenetic analyses of languages are consequently limited to estimating the tree and the divergence times between languages. The evolutionary dynamics of the sounds of language can be investigated only with difficulty.

Third, the transformation of multistate cognate relationships into binary characters introduces dependencies into the data, in that the values of the binary characters are not independent of one another. For a given concept, if a language has a word belonging to one cognate class, it will not have words from other cognate classes (unless of course polymorphisms are included). The continuous-time Markov chains that model the evolution of lexical items in cognate classes assume that character evolution is independent and identically distributed (i.i.d.). Binary characters derived from multistate cognate data violate this assumption.

Finally, cognate-class designations such as those in Table 1 serve as the observations for phylogenetic inference even though they are not observed. Cognacy judgments are hypotheses about ancestral relationships among words from related languages and cannot be observed any more than a phylogenetic tree can. The use of root-meaning traits is thus at odds with the fundamental principles of Bayesian inference, according to which the posterior distributions are conditioned on observed data. Our approach partially overcomes this inconsistency. Inferences are drawn from observable word forms and not abstract cognate relationships. At the same time, the input word-forms are selected on the basis of assumed cognacy.

1.2. The TKF91 approach

On account of these challenges, we introduce an approach to linguistic phylogenetics in which the observations are cognate word forms (Bouchard-Côté et al. 2013; Abner et al. 2024). The segmental history of the word forms is modeled with a continuous-time Markov model that allows one of three types of events to occur in an instant of time: (i) a transition from one segment to another; (ii) the



Fig 1. Approximate geographic distribution of the Romance languages examined in this study.

insertion of a single segment; or (iii) the deletion of a segment. Our treatment allows us to analyze data in a manner akin to molecular phylogenetic analyses in biology. This model was first introduced by Thorne et al. (1991), and is known as TKF91 in the molecular evolution literature. Just as nucleotides are considered equivalent states regardless of their position in the genome, our method treats segments as equivalent across words. We can therefore learn about the rates of individual event types by pooling information across different words (or across different segments of a word). This approach circumvents the four problems of root-meaning traits described above.

Although the TKF91 model can in principle be used to infer phylogenies, we focus in this study on its ability to estimate rates of change among segments, as it is of central importance to both linguistic theory and linguistic typology. There are a variety of methods for estimating the diachronic stability of segments (Wichman 2009; Moran et al. 2018; Moran et al. 2021), but none is phylogenetically based or uses models of segmental transitions. Indeed, one of the strengths of our approach is that it can take advantage of a rich set of segment-transition models, which are not limited by considerations of the labeling assigned to the states. We illustrate the power of the TKF91 model with a sample of eight Romance languages plus Latin, on the basis of which we find robust evidence for the following three claims regarding sound change between Latin and Romance. First, vowels are for the most part less stable over time than consonants. Second, a model of sound change in which parameters are shared among natural classes of phonemes outperforms simpler models. Finally, the equilibrium frequencies indicate that Romance languages make disproportionately frequent use of stops and short vowels in word forms, relative to other segment classes.

The remainder of this paper is structured as follows. Sections 2 and 3 introduce the data and methods, respectively, which are followed by a presentation of the results in Section 4. Section 5 situates these results within historical phonology and phonological typology and compares our approach to the other views of sound change. Section 6 brings the paper to a close with a recapitulation of the main points and identifies possibilities for expanding the TKF91 approach.

2. Data

 This study is based on a dataset of cognate word forms from a sample of eight Romance languages and Latin, [an assortment of languages which are all typologically fusional](#), whose geographic distribution is presented in Figure 1. The word forms are cognate in the traditional sense:

Table 2. Manual alignment of the words for the concept ‘what.’

Language	Phonemic Representation	Alignment
Latin	/kʷid/	kʷ id
Romanian	/tʃe/	tʃ e
Portuguese	/ki/	k i
Spanish	/ke/	k e
Catalan	/kɛ/	k ɛ
French	/kwa/	kwa
Walloon	/kwɛ/	kwɛ
Friulian	/tʃe/	tʃ e
Italian	/ke/	k e

they descend from a common ancestor (Chang et al. 2015, p. 201). In contrast to root-meaning traits (such as the one illustrated in Table 1), meaning is irrelevant to cognacy. For instance, the Latin adjective *gravis* means ‘heavy,’ but French *grave* means ‘serious.’ Despite this semantic difference, they are assigned to the same cognate set because they are segmentally homologous.

Cognate word forms were selected as follows. We first identified basic vocabulary items in Latin on the basis of the Swadesh 207 concept list. We could have selected cognates with any meanings, but used a Swadesh list solely to reduce the probability of encountering borrowed words. For each Latin lexeme, we collected its cognates in our eight Romance languages from etymological dictionaries. Forms thought to be due to borrowing either from Latin or from another Romance language were excluded. The orientation toward Latin is purely practical. We only used cognate sets in which a cognate could be identified for each language. By starting from Latin, we maximized the chances of having cognates in all of the Romance languages in our sample. [45 \(It is in principle possible to use incomplete cognate sets, i.e., sets in which at least one language lacks a cognate, but use of complete cognate sets simplifies parameter estimation, which is why we have followed this practice for this initial study.\)](#)

[23](#) In total, the dataset contains 102 cognate sets, which are comprised of 918 word forms containing ~~79 unique segments drawn from the following 98 concepts: all, and, at, belly, black, blood, bone, dog, dry, ear, earth, egg, eye, feather, fingernail, fire, five, flower, foot, four, fruit, full, good, grass, green, head, I, if, in, knee, leaf, liver, long, louse, meat, moon, mountain, mouth, narrow, near, new, night, not, old, one, other, rain, red, river, round, salt, sea, seed, skin, sky, star, stone, tail, there, thick, thin, three, come, cut, drink, fall, fight, flow, hear, hold, know, laugh, lie, asinabed, play, say, see, sew, sing, sleep, smell, squeeze, stand, swell, think, turn, wash, tongue, tooth, tree, warm, water, what, when, wind, woman, year, you (pl.), and you (sg.).~~ [unique segments drawn from 98 concepts.](#) The word forms can be represented as either phonetic or phonemic sequences. We use phonemic representations for two reasons. First, phonetic data is harder to come by, especially with a corpus language such as Latin. Second, the current implementation of the model is unable to take into account phonetic variation of word forms. Consequently, use of phonetic word forms would have required selecting a single representative. The representation of word forms is strictly segmental. Suprasegmental features such as stress and syllable structure are not represented. All segments are coded in standard Unicode IPA (International Phonetic Association 1999).

The word forms in each cognate set are manually aligned, an example of which is provided in Table 2 for the concept ‘what.’ Dashes (‘-’) denote the absence of a homologous segment present in other word forms. For example, the /i/ of Latin /k^w-id/ develops into /wa/ in French and Walloon. To

Table 3. Model: “Natural Class”

Long Vowel	e: o: u: a: i: ø:
Nasal Vowel	ẽw ẽ ẽ̃ ã ẽ̃ ɔ̃ õ ĩ ü œ̃ ẽj
Diphthong	qa ij iw ɔj aʊ ej ej̃ ɛa aj oj ej̃
Short Vowel	o i a ə ɐ ɔ e u i y ɛ œ ɑ ø
Nasal Consonant	m n ɲ ŋ
Liquid	l ʎ r ʀ ʁ
Approximant	j ɥ w
Affricate	tʃ ts dʒ
Fricative	ʃ x v f s h ʝ z ʒ θ ɣ
Stop	d k c p b t g g ^w k ^j k ^w

accommodate the emergence of /w/, there is an empty segmental slot before the vowel in all other languages. As described below in Section 3, however, manual alignments of cognate word forms serve only as a starting point for the Markov chain Monte Carlo (MCMC) sampling procedure, so our analysis does not condition on any particular alignment of segments. In other words, it is not necessary to include the dashes in the input alignments.

Latin is a highly synthetic language in which most lexical items having multiple word forms. In this study, each lexical item is represented by a single word-form, however. For verbs, the present active infinitive is used. For nouns, adjectives, and pronouns, singular forms in either the accusative or the nominative case are used, since these are the two case forms cognate with Romance word forms. Among nouns and adjectives in our dataset, the ancestral form is, *as expected*, far more often the accusative. The selection of word forms is not without consequences, since it can impact the estimated transition rates. For instance, most masculine singular nominative adjectives in Latin end in /-us/. By contrast, feminine singular nominative adjectives end in /-a/. Use of the feminine forms would increase the number of transitions originating in this vowel.

4 The lexical histories of the Romance languages are notoriously complex (Stefenelli 1992; Dworkin 2016), as they have been shaped not only by inheritance but also by borrowings from other Romance languages as well as Latin itself. Given this non-tree-like evolution, it is natural to question whether Romance cognates are appropriate for our phylogenetic method. In fact, Romance data are ideally suited to our goals because the histories of individual cognate sets are well understood and the risk of horizontal transmission in our dataset is low. Two reasons are responsible for this reduced risk. First, our dataset focuses on basic vocabulary, where lexical borrowing is less common (Tadmor et al. 2010; Carling et al. 2019, p. 2). Second, the phonological developments of the Romance languages, *whose earliest written testimonies go as far back as the 9th and 10th centuries, so that their history is exceptionally well known*, are sufficiently well understood that borrowings and Latinisms can often be detected. We excluded all word forms for which evidence of borrowing could be identified. Romance cognates are therefore appropriate for our analysis for the same reason that scholars have been able to reconstruct Proto-Romance lexemes using the traditional comparative method. For instance, the *Dictionnaire Étymologique Roman* (DÉRom 2008) reconstructs a subset of the Proto-Romance lexicon from a dataset similar in spirit to ours.

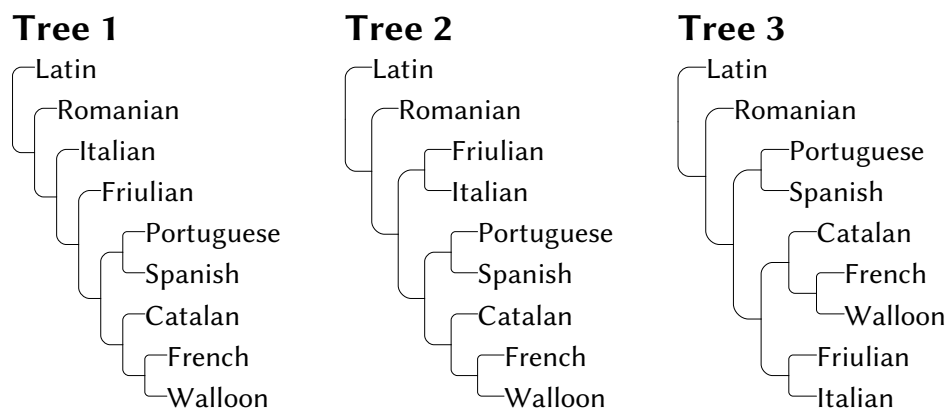


Fig 2. Model trees used in all analyses

3. Methods

A detailed description of the model and analysis can be found in the Supplemental Material. What follows is an outline, sufficient in detail to provide an understanding of the model and experiments performed in this study.

3.1. Model and Statistical Inference

We assume that the languages under investigation are related to one another through an unknown phylogenetic tree, $\Psi = (\tau, \nu)$, that contains information on the relatedness of the languages (τ) and the expected number of phonemic transitions ν that occur along each branch of the tree. Language evolves along the branches of the tree according to the TKF91 model (Thorne et al. 1991), which allows either a segmental transition, insertion, or deletion to occur in an instant of time. Insertions and deletions occur at rates λ and μ respectively, with $\lambda < \mu$. Phonemic substitution is modeled as a continuous-time Markov model with phonemes as the states of the process (as noted above, our inventory consists of 79 phonemes). The rates of change between all pairs of states are contained in the rate matrix $\mathbf{Q}$, which is governed by parameters θ .

We perform estimation in a Bayesian framework, in which parameter estimates are based on the joint posterior probability distribution:

$$f(\Psi, \lambda, \mu, \theta | \mathbf{S}) = \frac{f(\mathbf{S} | \Psi, \lambda, \mu, \theta) f(\Psi, \lambda, \mu, \theta)}{f(\mathbf{S})} \quad (1)$$

where $\mathbf{S}$ are the observed segments for the cognate words. The likelihood function, $f(\mathbf{S} | \Psi, \lambda, \mu, \theta)$, is calculated using the algorithm of Lunter *et al.* (Lunter et al. 2003). We use priors for the model parameters that are standard in phylogenetics, except for the rate parameters for the insertions and deletions, which are assumed to follow independent and identically distributed exponential distributions with $\lambda < \mu$. An alignment of length L receives a prior probability that is geometric with probability λ/μ . (Under the TKF91 model the length of an alignment at equilibrium is geometrically distributed with parameter λ/μ .) A more coherent alternative would be to define a fully generative prior on the alignment conditional on the insertion and deletion rates, building directly on the TKF91 process itself. Developing such priors for segmental homology structures remains an important

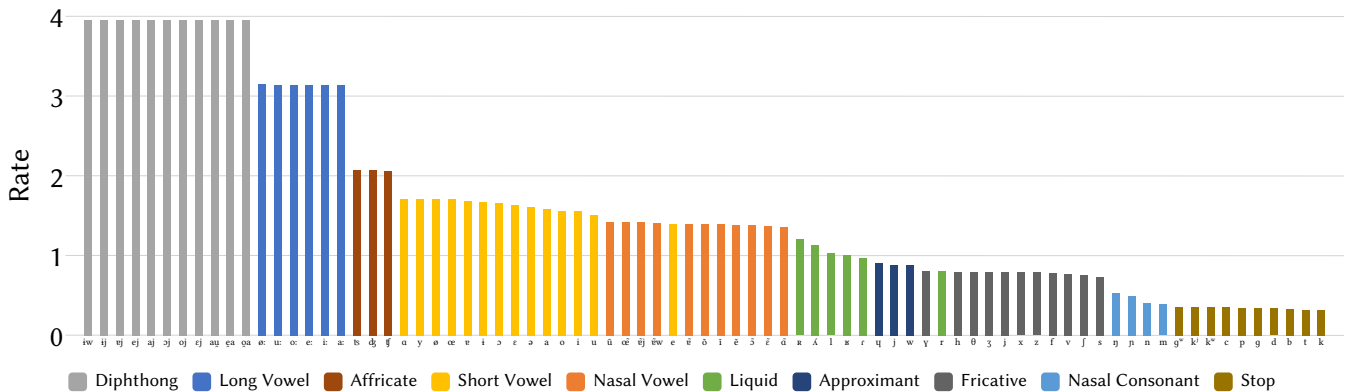


Fig 3. The rate of change when in a particular segmental state (or — q_{ii})

direction for future work.

We numerically approximate the posterior distribution of the parameters using MCMC. Specifically, we constructed a Markov chain that has as its states the parameters of the model and a stationary distribution that is the posterior distribution of interest. Samples from this chain when at stationarity are valid, albeit dependent, samples from the posterior distribution. Besides sampling the phylogenetic parameters of the model, the chain also samples segmental alignments (Lunter et al. 2004).

44 Our model of segmental transitions is based on phonological natural classes, which are classes of segments that share phonetic properties. The segmental inventory of our dataset was exhaustively partitioned into ten classes, which are presented in Table 3. This ‘Natural Class’ model allows different rates among the ten classes of segments. It has 79 parameters for the equilibrium distribution and 80 parameters describing rates among natural classes. The equilibrium frequencies are modeled as random variables drawn from a flat Dirichlet prior distribution.

3.2. Modeling linguistic history with cognate word-forms

The use of cognate words introduces several challenges, such as the alignment of homologous segments of cognate word forms. In a molecular phylogenetic analysis, the fine-scale homology of nucleotides sampled from different species is established using sequence alignment programs. However, even for molecular data in which long nucleotide sequences are the norm, there can be substantial uncertainty in the alignment (Wong et al. 2008). Alignment uncertainty is exacerbated for cognates because the word forms typically have only a handful of segments. We address this problem by marginalizing over segmental alignments. That is, we consider all possible word alignments, weighting each by its probability under the model. We do this by performing parameter estimation in a Bayesian framework, using Markov chain Monte Carlo, or MCMC (Metropolis et al. 1953; Hastings 1970), to sample model parameters, including segmental alignments, in proportion to their posterior probabilities.

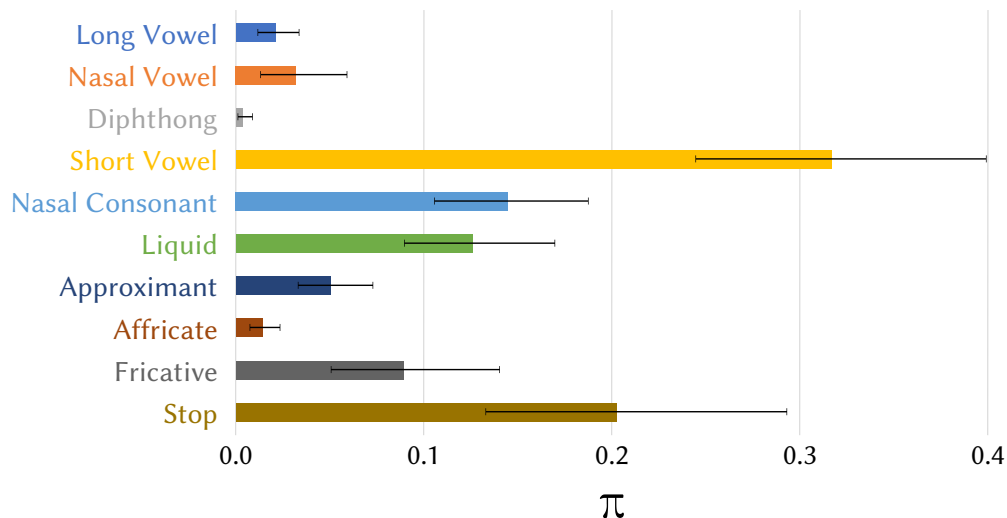


Fig 4. The equilibrium frequencies π of each natural class

3.3. Models of segmental change

We consider three models of segmental change in this study: M_1 , a model isomorphic to the model of DNA sequence evolution first described by Jukes and Cantor (Jukes et al. 1969) with equal rates between all pairs of states and in which the number of substitutions follows a Poisson distribution (hereafter referred to as the ‘Poisson’ model); M_2 , a model isomorphic to the model of DNA sequence evolution first described by Felsenstein (Felsenstein 1981) in which the rate of change between a pair of segments is proportional to the equilibrium frequency of the destination segment; and M_3 , the Natural Class model, in which segments that share articulatory properties are grouped together with different rates of change within and between these segmental classes. The number of estimated parameters varies across models. The first model, M_1 , has no free parameters. The second model, M_2 , has 78 free parameters (one less than the number of segments). The Natural Class model, M_3 , has 55 additional segmental class rate parameters: $78 + 55 = 133$ free parameters.

3.4. Data Curation and Analysis

Our analysis is coordinated using a program written in the Nytril programming language (McCreight et al. 2024). We created an automated framework for the experiment that comprises a reusable IPA library, word forms, data quality checks, typeset output, and input files for the MCMC software (written in C++).

Word forms from the nine languages in our study are organized into classes by concept and

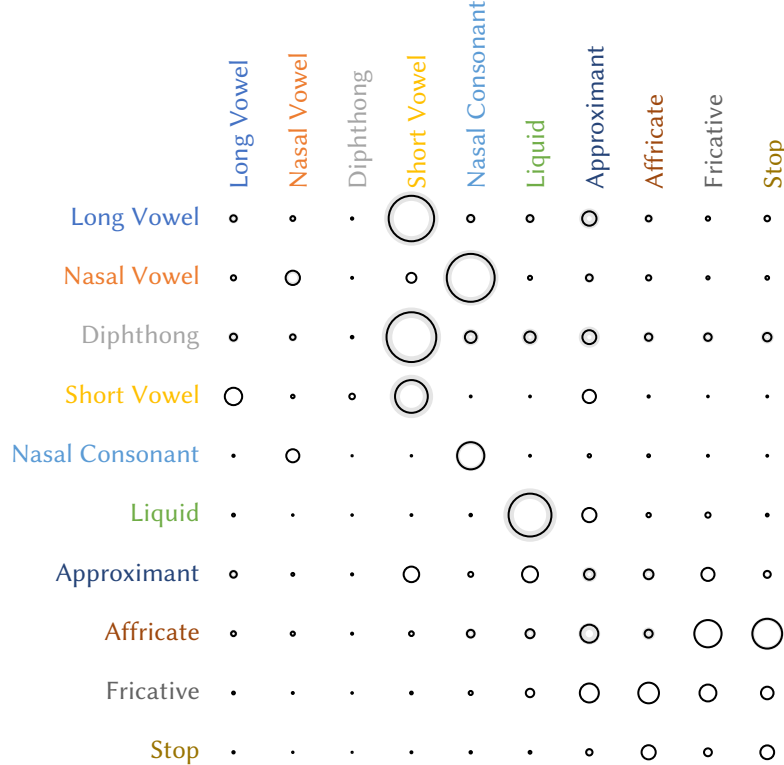


Fig 5. The average rates of change between and within the ten segment groups of the Natural Class model. Circle area is proportional to the rate of change between groups (q_{ij}). Shaded area represents the 95% credible interval.

then by cognate class. Figure 10 shows an example of the raw data for the concept ‘mountain.’ A challenge for programming in linguistics is that the IPA representation of words is expressed in Unicode (UCS), ~~not ASCII or ANSI. This requires care with file formats, text encoding, and diacritic compositional form, which cannot be represented in a single-byte encoding such as ASCII.~~ Instead, it is best stored in UTF-8 which is a common, cross-platform, multibyte encoding for UCS. Diacritic compositional form must be standardized to facilitate string equality testing. We selected Form C, but what matters is that the form is consistent across files. Furthermore, each IPA segment string can comprise several UCS code points, so that the mapping between string characters and segments is not one-to-one. Figure 11 presents an example of a more complex encoding where the numbers in grey are the hexadecimal UCS code points for the letters and diacritics. Since our analysis requires each word to be separated into segments, a parser converts word strings into arrays of segment objects that facilitate the manipulation of segments at a high level. A reusable library of such objects endows each segment with linguistic features that can be used for analysis. After the input step, all steps are coded using properties such as ‘vowel’ and ‘nasal,’ rather than working with strings of characters directly.

After the raw input is parsed, the control program aggregates and partitions the words and builds the data files for the MCMC program to perform Bayesian phylogenetic analysis under the TKF91 model. The MCMC analysis runs for 5,000,000 cycles and repeats to check for convergence. Samples taken during the first 15,000 cycles of the chain are discarded as burn-in. When the analysis is complete, the results are read back into the control program. The resulting tables, trees, and graphical

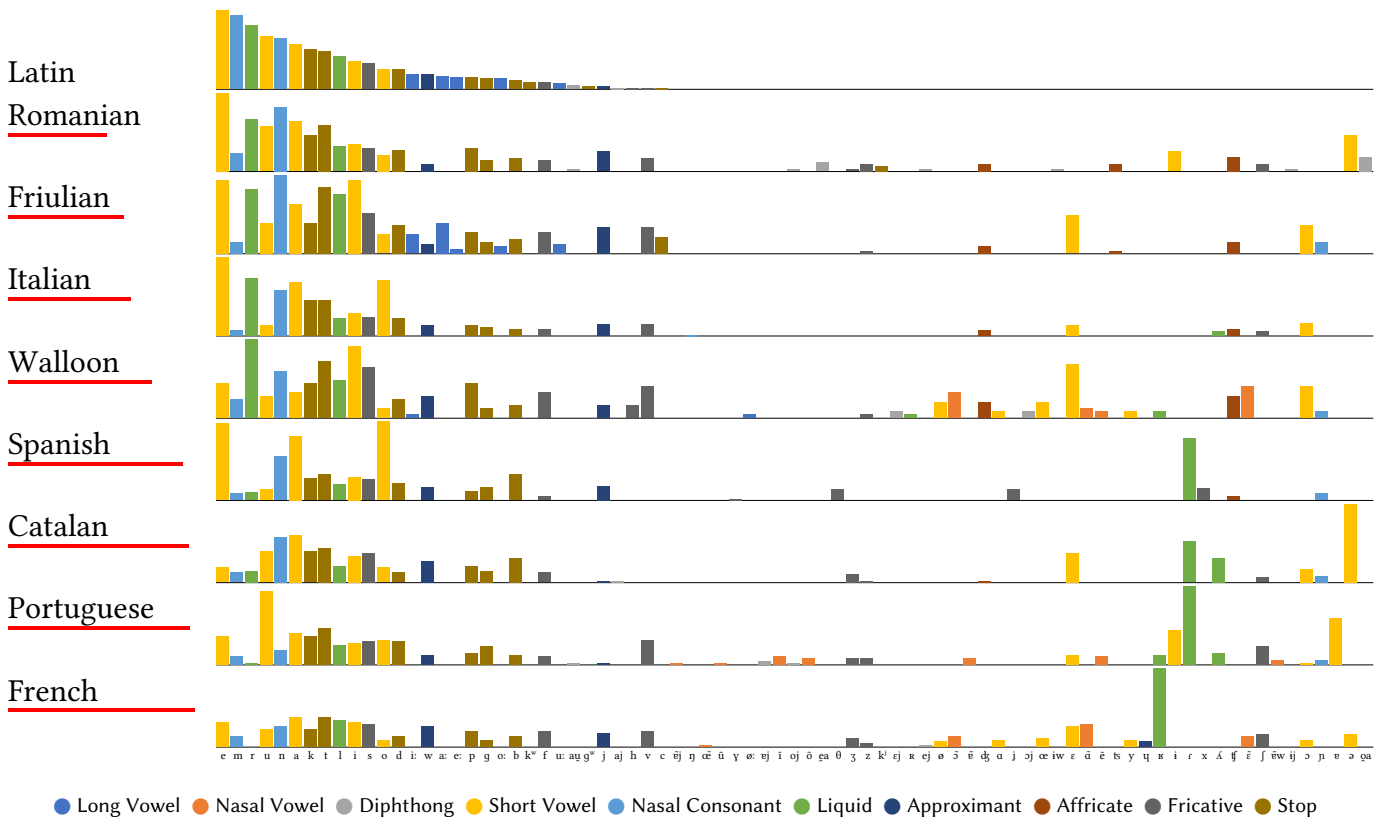


Fig 6. Comparison of occurrence rates of segments in each language. Segments are sorted by the rates in Latin. Bars are normalized to the rate of the most frequently occurring segment in each language. Languages are ordered by their Euclidian distance from Latin indicated by the red line.

figures are automatically generated from the MCMC output and combined with narrative elements created by the authors to make the final paper and supplemental materials.

4. Results

4.1. Phylogeny

Our statistical analysis treats all model parameters as random variables with prior probability distributions. The tree topology describing the relationships of the languages is also considered a random variable with a uniform prior probability in which all possible tree topologies are assigned equal prior probability. As described in the Methods section, we use MCMC to numerically approximate the posterior probability distribution of the model parameters. Our analyses consistently found a handful of topologies that best explained the data. Unfortunately, the MCMC poorly explored the tree space: the MCMC algorithm would find a good tree then become stuck on it.

In our MCMC implementation, parameters are updated in blocks. A parameter is randomly selected and then updated. In a single MCMC cycle, either the tree topology or the alignment is

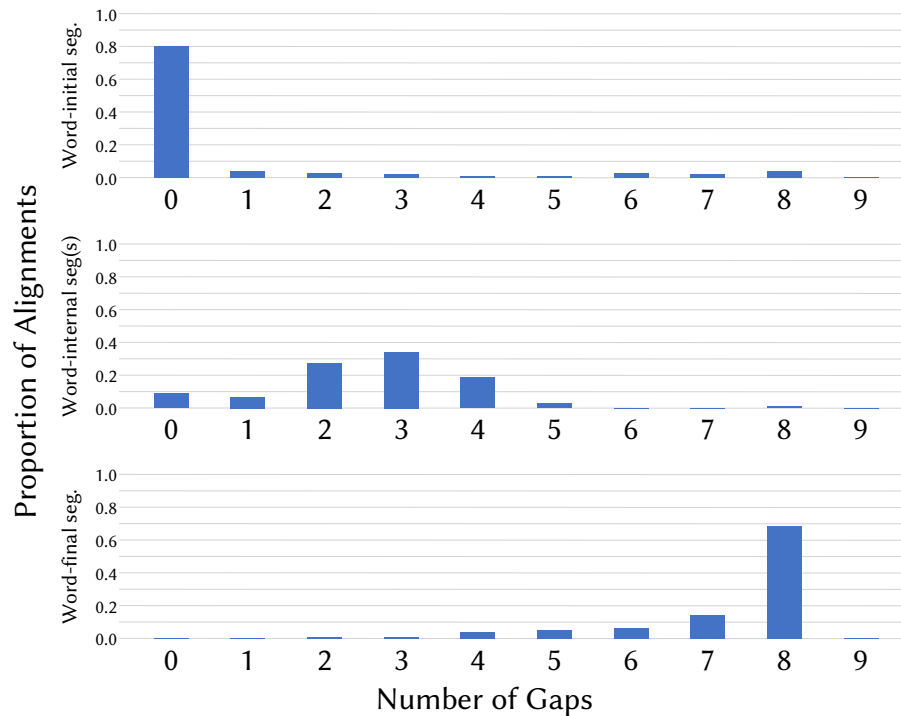


Fig 7. The number of gaps observed across the 102 cognate sets analyzed in this study at the word-initial segment (top), word-internal segment(s) (middle), and word-final segment (bottom). Nine languages were analyzed in this study, so it is not possible for there to be nine gaps. Deletion events that led to a gap for all the languages would not be included in an alignment, and is explicitly conditioned on in Lunter et al. (Lunter et al. 2003). There is a clear, and statistically significant, pattern of an increasing number of gaps as one moves from the initial to the final segment of the word. (A likelihood ratio test of the null hypothesis that the distributions are the same at the initial, internal, and final segments is highly significant, with a test statistic of 425.62; $d.f. = 16$, $p < 0.001$.)

updated, but not both simultaneously. The tree topology and alignments are confounded. As the MCMC procedure found better trees, the alignments that were accepted by the MCMC procedure tended to fit the current tree in computer memory. As the alignment becomes better adapted to the current tree in memory, it becomes more difficult for the MCMC procedure to accept alternative topologies, even if they would be sensible on a different alignment.

This problem was likely not encountered in the other implementations of the TKF91 model (Lunter et al. 2004), which analyzed an alignment of amino acid sequences. In our analysis of cognate data, the insertion and deletion rates were estimated to be quite high relative to the substitution rate: $\lambda = 0.243$ (0.217, 0.273) and $\mu = 0.298$ (0.266, 0.335) for the insertion and deletion rates, respectively. (The credible intervals in the parentheses contain the true value of the parameter with probability 0.950.) Roughly speaking, this means that for every two segmental transitions there was one insertion or deletion event. Just like substitution events, insertion and deletion events can be informative about phylogeny. Unlike the earlier analysis of amino acid sequence data, the insertion and deletion events were strongly informative about specific tree topologies, and this tree support could change depending on the homology assignments for the alignment.

In all subsequent analyses, we use the three topologies presented in Figure 2. Although there is considerable debate about the topology of the Romance languages, in recent decades most specialists have converged on the view that the first two clades to form are Sardinian followed by Romanian

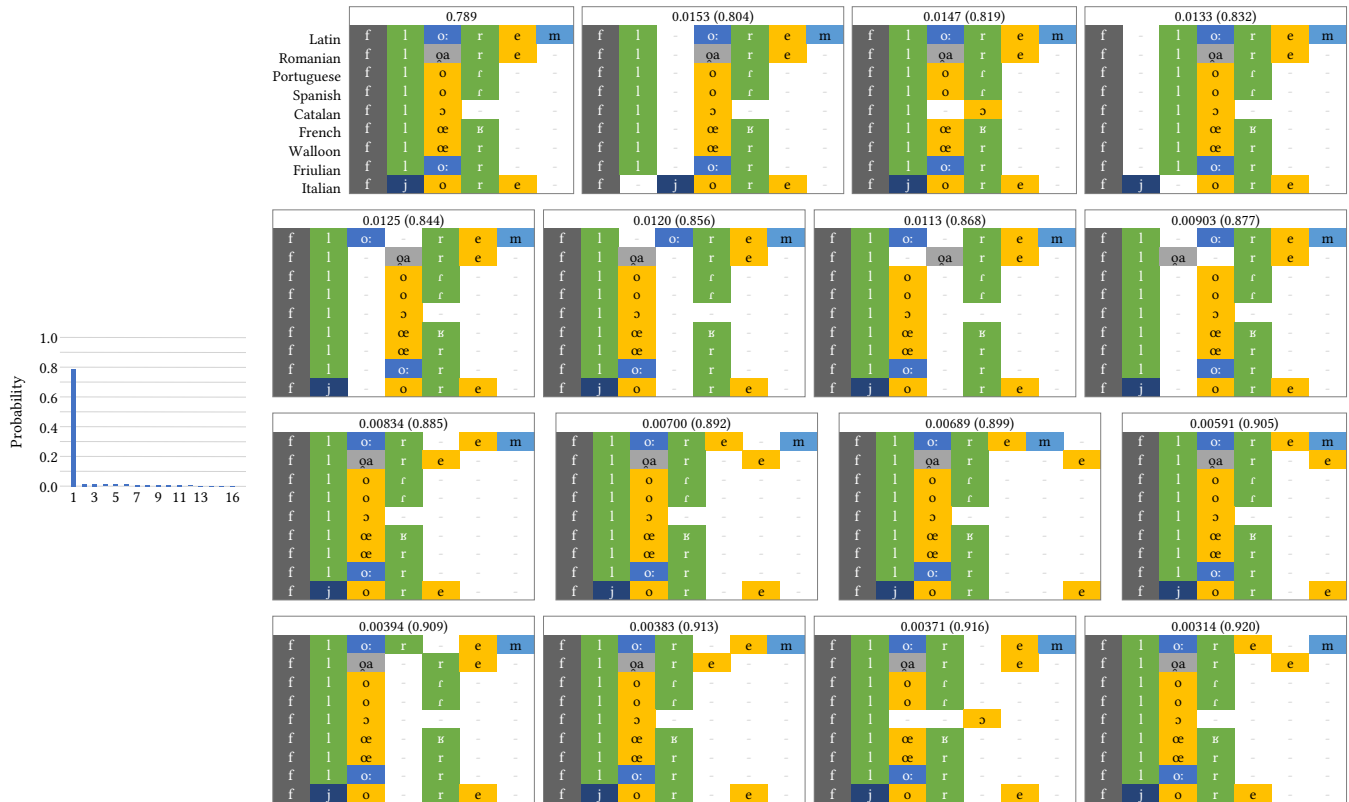


Fig 8. Segments are colored according to the Natural Class model. Dashes denote gaps where there is no homologous segment for the alignment position. The alignments are ordered from the highest posterior probability (top-left alignment) to the lowest posterior probability (bottom-right alignment) in row order.

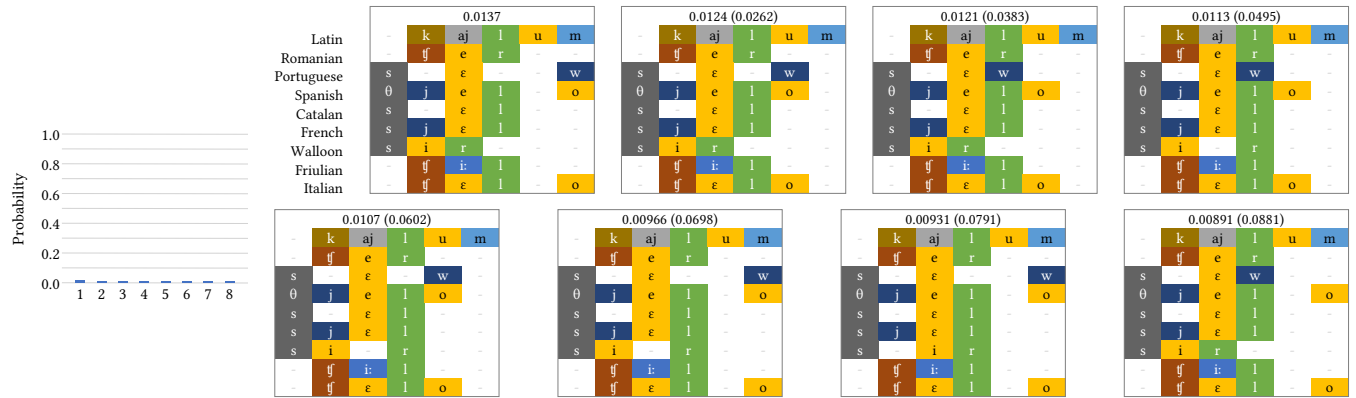
(Hall 1950; Hall 1976; De Dardel 1985; Swiggers 2001; Vallejo 2012; Buchi et al. 2015; Dworkin 2016). Since Sardinian is not in our study group, the clade containing Romanian is the first to form in each of the trees in Figure 2. The sub-trees containing the remaining dialects differ in their topologies, but they all contain Spanish and Portuguese as a clade, as well as Catalan, French, and Walloon.

There is extensive debate concerning the relationship between Classical Latin, Vulgar Latin, [Proto-Romance](#), and the Romance languages (Posner 1996; Harris et al. 2003; Chang et al. 2015; Heggarty et al. 2023; Goldstein 2024). The central question is whether Latin (be it Classical or Vulgar) is a sampled ancestor of the Romance languages or their sibling. In this study, we adopt the latter assumption and treat Latin as an outgroup to the Romance languages. This choice is motivated by practical rather than linguistic considerations, since phylogenetic models that allow for sampled ancestry are substantially more complex than the tree model employed here. Importantly, however, our treatment of Latin does not preclude an ancestral interpretation: if the estimated branch length leading to Latin is zero, Latin is effectively placed at the root of the Romance clade.

4.2. Model Comparison

We compared the fit of the three models described above in section 3 to the data using the Bayes

M_1 – Poisson



M_3 – Natural Class

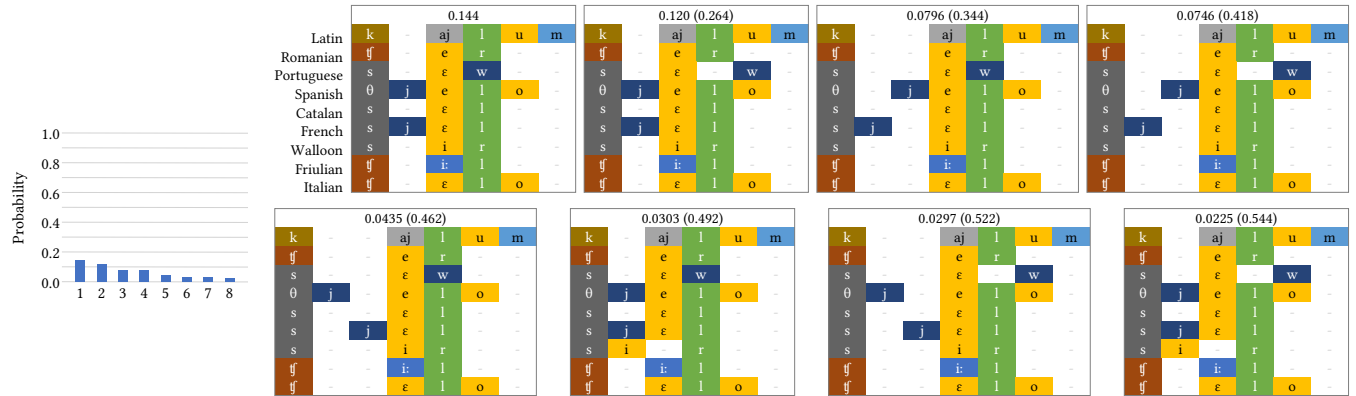


Fig 9. Alignments of highest posterior probability of the cognate for the concept ‘sky.’ The eight alignments under the M_1 (Poisson model) account for 0.0881 posterior probability. The eight alignments under the M_3 (Natural Class model) account for 0.544 posterior probability.

factor. The Bayes factor is the ratio of the marginal likelihoods,

$$BF = \frac{f(\mathbf{S}|M_i)}{f(\mathbf{S}|M_j)} \quad (2)$$

with ratios greater than one favoring model M_i . As the models examined are nested, we could calculate the Bayes factor using the Savage-Dickey ratio (Dickey 1971), which is the ratio of the posterior to prior densities evaluated at the restriction that makes the parameter-rich model equivalent to the simpler model. The Natural Class model explains the data substantially better than the two simpler models, with log Bayes factors of $\ln BF_{12} = -1074.2$ and $\ln BF_{23} = -117.3$, respectively, for a comparison of M_1 to M_2 and M_2 to M_3 . This indicates overwhelming support for the more parameter-rich Natural Class model (Jeffreys 1939).

4.3. Segmental Change

Figure 3 presents the absolute values of the diagonal elements of the rate matrix of the continuous-time Markov model describing phonemic change estimated under the Natural Class model. These diagonal elements can be interpreted as a measure of segmental volatility, since they are the sum of the rates at which they transition to other segments. When the process is in segment i , $-q_{ii}$ is the rate at which it changes to another segment. The higher the rate, the more rapidly a change to another segment occurs on average. The rates for several segmental classes in Figure 3 appear to be identical (e.g., diphthongs, long vowels, affricates, and nasal vowels, but this is an artefact of the visualization. The rates in these classes are not identical, but they are so small that they are not visible in this plot.

The most striking aspect of Figure 3 is the division between vowels and consonants. With the exception of affricates, vowels are the most volatile segments and within this class diphthongs and long vowels are the least stable. Overall, the most stable segments are all consonants, with the stops the most stable of all. The rates of change are also remarkably similar within each class: all the long vowels cluster together, as do the affricates, fricatives, short vowels, nasal vowels, fricatives, nasal consonants, and stops. The only exceptions are the liquid /r/, which appears with the fricatives, and the mid-front vowel /e/, which appears among the nasal vowels. This clustering is driven by the exchangeability rates that are shared between classes of segments and the degree of variation in the equilibrium frequencies. When there the equilibrium frequencies of the segments in a given class vary less, the $-q_{ii}$ values of the segments in that class exhibit greater similarity. [42 The clustering in](#)

```
namespace Mountain {
  namespace Primary {
    Latin      = "montem";
    Romanian   = "munte-";
    Portuguese = "mõ-ti-";
    Spanish    = "monte-";
    Catalan    = "mun---";
    French     = "mɔ̃----";
    Walloon    = "mɔ̃----";
    Friulian   = "mont--";
    Italian    = "monte-";
  }
}
```

Fig 10. The concept ‘mountain’ as coded in the data file.

Figure 3 thus reflects both the model structure and empirical patterns in the data.

Figure 4 shows the estimated equilibrium frequencies of the Natural Class model M_3 . These equilibrium frequencies play two roles in the calculation of the likelihood of the observed data. First, they define the prior probability distribution over segmental states at the root of the tree. Second, for models M_2 and M_3 , they modulate the rates of change between segments, such that the rate of transition is proportional to the equilibrium frequency of the destination segment. Consistent with this formulation, transitions into liquids (0.188), short vowels (0.110), and nasal consonants (0.0758) have the highest estimated rates, corresponding to their elevated equilibrium frequencies. More generally, the estimated equilibrium frequencies closely track the empirical frequencies of segments in the data.

Figure 5 shows the estimated rates of change between the 10 groups of the Natural Class model. Rates of change are generally highest among segments in the same class. The estimated rates of change from one segment to another under the Natural Class model are 0.33, 1.63, 0.05, 8.59, 5.91, 14.70, 0.90, 0.52, 2.28, and 1.52 times higher than expected under an equal rates Poisson model for long vowels, nasal vowels, diphthongs, short vowels, nasal consonants, liquids, approximants, affricates, fricatives, and stops, respectively. (In other words, the long vowel $\rightarrow$ long vowel, diphthong $\rightarrow$ diphthong, approximant $\rightarrow$ approximant, and affricate $\rightarrow$ affricate rates were lower than would be expected if all rates of change were the same.) Long vowels and diphthongs counter the general pattern as they transition more often to short vowels than to long vowels or diphthongs, a change we discuss further below. Note that 60.0% (6 out of 10) of the changes within a class are higher than expected under an equal rates model, whereas 23.3% (21 out of 90) of the changes between different natural classes are higher than expected under an equal rates model, supporting the pattern of a generally higher rate of change between segments assigned to the same natural class.

4.4. Segmental Alignments

Our method treats the homology relationships of segments between languages as a random variable. We summarize the posterior distribution by constructing a credible set of alignments, where a credible set of alignments is constructed by ordering the alignments for a cognate from the highest to lowest posterior probability. The alignments are then included in a credible set, starting with the alignment with the highest posterior probability, until the cumulative posterior probability is 0.95. Figure 8 shows alignments for the concept ‘flower’, which illustrates three key properties of all posterior alignments. First, the distributions of credible sets are heavily weighted in favor of a single alignment with most of the posterior probability. In this case, one alignment had a posterior probability of 0.79. Second, insertions and deletions occur predominantly at word-end, where gaps are more frequent on average compared to other positions (Figure 7). Finally, the alignments tend to assign segments from the same class to the same column in the alignment.

Interestingly, the distribution of alignments depends on the model of segmental change. Figure 9 shows the 95% credible set of alignments for the concept ‘sky’ under the Poisson and Natural

t̪ʰ

0074, 0279, 031d, 030a

Alveolar Pulmonic Affricate

Fig 11. The IPA-Unicode encoding of a segment.

Class models. The credible set contains 1363 alignments under the Poisson model, but only 226 alignments under the Natural Class model. This difference makes sense given the rates estimated under the Natural Class (Figure 5). For example, the rate of change between /u/ and /o/ are estimated to be high, whereas the rate between consonants and /o/ is low. The distribution of alignments under the Poisson often makes consonants homologous with /o/. Under the Natural Class model, however, the same alignments are improbable, as the rates of substitution between those alignments are low.

5. Discussion

5.1. The TKF91 Framework Compared to Other Approaches

Approaches to sound change in the linguistics literature vary. One prominent view identifies phonemes as the basic units of change (Bloomfield 1933), according to which sound change is abrupt and non-independent. It is abrupt in that there are no intermediary stages in a transition from /a/ to /o/, for instance. It is non-independent since the change from one phoneme to another across all lexical items is regarded as a single event. Other scholars emphasize the role of acoustic and auditory phonetics in sound change (Ohala 2003; Blevins 2004; Ohala 2012), while yet another view contends that sound change occurs by gradually making its way through the lexicon in a process known as lexical diffusion (Chen et al. 1975; Phillips et al. 2015).

Our framework is not situated within any one of these traditions, but rather has affinities with the phonemic approach and lexical diffusion. Since our observations are phonemic strings, it is phonemes that undergo insertions, deletions, and transitions. In contrast to the standard assumption of historical linguistics, however, these changes are treated as independent. For instance, a change from /a/ to /o/ is not modeled as a single transition across all word-forms in our study. Instead, a rate of change is inferred based on the homologies of /a/ and /o/ in the data and how often the former transitions to the latter.

The application of the TKF91 model to linguistic data is similar to the analysis of Bouchard-Côté et al. (Bouchard-Côté et al. 2013), who also treat cognate word-forms as observations. Their approach differs, however, in that they applied a string transducer model (Holmes et al. 2001) to characterize segmental transitions. The transducer model has the advantage that it can account for many of the processes that are known to influence segmental change, such as context dependence. Our method, as currently implemented, does not consider the environment of segmental changes. However, unlike the transducer model, the TKF91 model is a historical model with well-defined events that can transform a word. In fact, the likelihood accounts for all the histories of insertion, deletion, and segmental substitution that could have led to the observed words sampled from the nine languages. These events occur at rates that can be estimated. Since the TKF91 model is event-based, it is a natural starting point for modeling sound change even if that starting point does not account for processes that are accommodated for in other models, such as the string transducer model.

 29 The complexity of the likelihood calculations under the TKF91 model is exponential in the number of taxa and word length. Specifically, the complexity is of order $O(2^N \bar{L}^N)$ where N is the number of leaves on the tree and $\bar{L}$ is the geometric mean of the word lengths (Lunter et al. 2003). The application of the TKF91 model to linguistic data is not as problematic as it is for DNA sequence data because words are much shorter than DNA sequences, with $1 \leq L \leq 10$. That said, one can obtain likelihood calculations that are linear in the number of tips if one assumes that the insertion rate is

independent of the word length, instead of the overall insertion rate being $\lambda(L + 1)$ as it is under the TKF91 model (Bouchard-Côté et al. 2013). This model—the Poisson Indel Process (PIP)—could be applied to linguistic data sets that are composed of many more languages than we analyzed here. Since the alignments visited by the MCMC procedure are similar in length, the PIP model might be a sensible option for linguistic data.

5.2. Insights into Sound Change

In contrast to traditional historical phonology, which is primarily concerned with individual sound changes and their relative chronologies, the TKF91 approach yields insights into broader systemic trends in the evolution of sounds and segmental inventories. We focus here on three measures that illuminate both aspects: the diagonal of the Q matrix, transition rates within and between natural classes, and equilibrium frequencies. Although a number of quantitative measures of sound change have been proposed in recent years, these measures constitute a novel contribution to the field.

As noted above in section 3, the diagonal of the Q matrix represents the rate at which a given segment transitions to any other segment and therefore provides a measure of its diachronic stability. The clear division between vowels and consonants in Figure 3 is consistent with results from previous research in Romance historical phonology. One of the major sound changes distinguishing Latin from the Romance languages is the loss of phonemic vowel length (Janson 1979; Loporcaro 2015). Likewise, the diphthongs of Classical Latin begin to undergo monophthongization already within Latin itself (Posner 1996, p. 106). Our results are also broadly consistent with the findings of Moran et al. (2021), pp. 92-95, who show that vowel inventories have changed more rapidly than consonant inventories in Indo-European.

Our findings also overlap to some extent with work in phonological typology that has sought to identify core sound inventories across the world’s languages. For example, the basic consonant inventory proposed by Nikolaev and Grossman (Nikolaev et al. 2020), /p t k n l r/, aligns with the trends we identify in Romance, insofar as the most stable classes in our analysis are stops and nasals. There is even stronger convergence with the set of primal consonants proposed by Bybee and Easterday (Bybee et al. 2022), /p t k b d g m n ŋ s l/. There are, however, two important differences between their notion of primal consonants and diachronic stability as measured by the diagonal of the Q matrix. First, these consonants are considered primal because they are only rarely created through sound change. That is, they infrequently arise as the output of sound change processes. Under our framework, by contrast, segmental stability is associated with the input to sound change, insofar as the diagonal of the Q matrix reflects the rate of transition *away* from a given phoneme. Second, both the basic and primal consonant inventories are intended to represent a core set of universal consonants across the world’s languages. While our results are relevant to the typology of diachronic phonology in that they capture broad tendencies in segmental evolution, they should not be interpreted as universal laws of sound change.

The average transition rates in Figure 5 profile a different aspect of segmental stability, as they refer to specific classes of inputs and outputs. In addition, the focus is not on transitions among individual segments, but rather transitions within and between natural classes. For instance, the circles on the diagonal represent the average transition rate within each natural class. The liquids exhibit the highest rate and the diphthongs the lowest. A liquid is thus on average more prone to transition to another liquid than to a segment from any other class. At the opposite end of the spectrum, when a diphthong changes it is least likely to transition to another diphthong. This trend is

due at least in part to liquid dissimilation, as in the change from Latin *arbor* to Spanish *árbol* (Abrego-Collier 2013; Müller 2013). Looking across the row for diphthongs in Figure 5, the highest rate of change is with the short vowels. These rates are driven by two common sound changes in the history of Latin and Romance, namely vowel breaking, whereby a monophthong becomes a diphthong, and monophthongization of diphthongs. The highest off-diagonal rates in Figure 5 all involve vowels, which is in line with the results discussed above. In addition to diphthongs and short vowels, the highest rates are also found among short vowels and long vowels as well as nasal vowels and nasal consonants. This latter trend is driven by the emergence of nasalized vowels in French and Portuguese.

The equilibrium frequencies presented in Figure 4 provide insight into the distribution of phonemic segments across major segment classes in Romance languages. The combined frequency of short vowels and stop consonants exceeds 0.5, indicating that on average more than half of the segments in a word are drawn from these two classes. This distributional skew reflects the diachronic developments discussed above. In particular, the reduction of diphthongs and long vowels leads to a decrease in their frequency of occurrence, with a corresponding increase in the frequency of short vowels.

Finally, the results of model comparison support the relevance of natural classes in sound change (King 1969, p. 201). It is well established that sound changes often target natural classes of segments. Grimm’s Law, for instance, systematically affects all [Proto-Germanic reflexes of](#) Proto-Indo-European stops—voiceless, voiced, and voiced aspirated. The regularity and non-randomness of the segments affected by such changes are due ultimately their phonetic grounding, as natural classes are groupings of sounds that pattern together in phonological processes precisely because of shared articulatory and acoustic features.

5.3. Segmental Sampling

Under our framework a new issue in linguistic phylogenetics emerges, segmental sampling (Dockum et al. 2019). Certain segmental transitions can play a crucial role in topological inference, but under a sampling scheme such as ours, in which selection of concepts is guided by a Swadesh list, there is no guarantee that the necessary segmental correspondences will be sampled. For instance, Romanian merges Proto-Romance **/ɔ/* and **/o/* to */o/* and **/ʊ/* and **/u/* to */u/* (in terms of classical Latin, it neutralizes vowel length distinctions). By contrast, Italo-Western Proto-Romance (i.e., the common ancestor of all other languages in our sample) merges **/ʊ/* and **/o/* to */o/* (in terms of classical Latin, it merged short <u> and long <o>). As a result, Latin *buccam* ‘mouth’ and Romanian *bucă* preserve */u/* against the */o/* of, for instance, Italian *bocca*. This correspondence is a key piece of evidence for the topological position of Romanian (Janson 1979, p. 25; Herman 2000, pp. 32-34) and if it is not sufficiently represented in the dataset it may not be possible to infer its place in the tree accurately. There is the further question of how representative the frequency distributions of segments, which are presented in Figure 6, are in the languages in our sample. These distributions are based on sets of word-forms sampled according to specific criteria and it is unclear how much they diverge from the true frequency distribution of each language. This potential divergence is important because it can impact the estimates of the transition rates.

5.4. Limitations of the event-based model

Our results are based on a model of segmental transition in which the three possible events described

above are applicable only to individual segments. Certain patterns of sound change cannot therefore be captured under such a model. For instance, metathesis, a change in which two segments exchange positions, is modeled as two separate transitions. Friulian /tarɔnt/ ultimately descends from Latin /rotundum/ ‘round,’ with the initial sequence /rVt/ of Latin having metathesized to /tVr/. Under the TKF91 model, this single event becomes two, with one change from /r/ to /t/ and another from /t/ to /r/ (leaving aside the possibility of intermediate transitions). A second limitation is that the context in which a segmental transition takes place plays no role in the estimation of transition rates. It is well known that the evolution of phonemes can be sensitive to neighboring sounds. Finally, there is currently no straightforward way to incorporate supra-segmental features such as stress, although it plays a crucial role in the phonological history of the Romance languages.

5.5. Advantages of the event-based approach

11 Our event-based approach to linguistic history has the power to transform the discipline by offering an inroad into questions that were previously intractable. For example, questions related to the relationship between phonological change and the phonemic inventory of a language can now be addressed. Does the frequency of a phoneme affect its diachronic stability? To what extent is phonemic change sensitive to the size and structure of the phonemic inventory of a language? What is the role of natural classes in sound change? This last question can be addressed through the comparison of models with different numbers of rate parameters.

By casting the problem of language change in a statistical framework, we inherit the benefits of the statistical approach to estimate model parameters and, perhaps more importantly, to compare alternative models of language change to find the model that best explains the data. In this way, linguists can learn about language evolution by devising models that account for potential linguistic phenomena, then testing the model against models that do not account for the linguistic feature. This approach has proven successful in molecular evolution where models have been described and tested that account for many evolutionary processes of molecular change.

6. Conclusion

24 We have presented the first application of the TKF91 model to a linguistic dataset and argued that an event-based approach to language evolution offers significant advantages over current practices. We believe the method can be extended to handle more complex segmental changes (such as those involving context dependency). Moreover, the event-based approach has the potential to improve divergence-time ~~estimation~~estimation, which relies on estimating the number of events that occurred along the branches of the tree. When applied to abstract cognate relationships, it is not clear what exactly is being counted along the branches. By contrast, the possible events in our event-based approach are well defined: segmental transitions, insertions, or deletions.

The code and data for this study are available on [GitHub](#).

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Comments and Responses

1) Reviewer 1

Comment

In this article the author propose to model the phonological evolution of word forms through time, and to infer jointly the phonological changes and the phylogeny of the tree (actually it seems the authors only select the topology from 3 proposed topologies). They propose several models and manage to compute a Bayes factor, which is a very good addition to the paper in my opinion. The main positive point of the model in my opinion is that the alignment of the word forms is automatic and a side produce of the inference. The main negative point in my opinion is the lack of importance given to H: depending on the choice of the prior on this parameter we can get extremely different results.

Response

We thank the reviewer for this thoughtful assessment of the paper. We are especially pleased that they view the automatic inference of segmental alignments as a major strength of the framework, and that the use of Bayes factors for model comparison is seen as a valuable contribution.

We fully agree that the homology structure H , the alignment of word segments across languages, is a central latent variable in the model and carries a great deal of linguistic information. In the present work, we have focused primarily on using H as a vehicle for defining a likelihood under a fully historical model of word evolution. However, we view the inferred posterior distribution over alignments as an important object of study in its own right.

For example, one could define probabilistic notions of segmental homology, such as the posterior probability that two segments in two languages descend from a common ancestral segment, or identify regions of words that exhibit strong versus weak homological support across languages. These are precisely the kinds of questions that linguists traditionally address qualitatively, and our framework provides a principled statistical way to approach them.

We also agree that the choice of prior on H is consequential. Because H defines the correspondence structure between words, different prior assumptions can influence the balance between insertions, deletions, and substitutions, and thereby affect downstream inferences. For this reason, we believe that the specification of priors on homology structures deserves careful theoretical attention.

In the present study, our goal was not to exhaustively explore this space of priors, but rather to demonstrate that fully historical, segment-level evolutionary models can be applied directly to linguistic data within a Bayesian phylogenetic framework.

We therefore view this paper as a proof of concept. It establishes a solid statistical foundation on which richer linguistic models, more expressive priors on homology, and more refined questions about sound change and lexical evolution can be built.

By casting word evolution in an event-based probabilistic framework, we aim to provide a platform that allows linguists to devise and test explicit evolutionary hypotheses, rather than relying on heuristic alignments or abstract cognate encodings.

In this sense, the role of H is not a weakness of the approach but one of its main strengths. It makes explicit the homology assumptions that are usually implicit, and it provides a natural locus for future methodological development.

2) Reviewer 1

Comment

Two articles come to my mind as comparison points for this work: the first is Bouchard-Côté's one (2013) that is cited but not enough commented in my opinion. The current article uses TKF91 as a base for their inference, while Bouchard-Côté's comments on this model that it lacks expressiveness, and I would agree with this comment. Obviously, as Bouchard-Côté mostly wants to infer the ancestral states, it is possible to use a more complex model.

Response

See our response to 13) below.

3) Reviewer 1

Comment

The second article would be Abner (2024) "Computational phylogenetics reveal histories of sign languages", which is closer in the goal but uses different assumptions on the model.

Response

We thank the reviewer for bringing this work to our attention and have added a reference to this paper in the manuscript. While the goal of the model in this paper is very close to our own, the actual model is quite different. To name just two non-trivial differences, it relies on fixed alignments and does not model insertions and deletions as such. In the paper, the model is actually not applied to cognate word-forms, but rather contrastive units of sign languages.

4) Reviewer 1

Comment

I don't think a direct comparison with any of those model is doable or interesting (although you mention that TKF91 is not primarily used to infer the phylogeny and thus Bouchard-Côté is your direct competitor), but I would think some more questions can be asked based on those articles (I am not a specialist of Romance languages so some remarks might just mean additional details on the specificities of this domain is needed). I mention those questions in the next parts.

Response

This comment is addressed in subsequent responses.

5) Reviewer 1

Comment

I think the article would benefit from more care in the writing. Some figures are not referred in the text (e.g. fig 11). For more clarity, a small table could summarize models M1, M2 and M3. These models could also be mentioned earlier. It is not entirely clear to me how are the parameters of M3 designed, it looks to me as a type of hierarchical model.

Response [View changes](#)

We thank the reviewer for this suggestion and agree that clarity of exposition is important, especially given the technical nature of the models.

Regarding the figures, Figure 11 is in fact referenced in the main text in Section 4.4, paragraph

2, where it is discussed in the context of model comparison and alignment behavior. We will nevertheless revise the manuscript to make this reference more explicit and easier to locate.

We agree that a summary table of models M1, M2, and M3 would improve readability. We will add a small table summarizing the structure and assumptions of each model and will introduce the models earlier in the manuscript to provide additional context before their formal definition.

With regard to model M3, it is not a hierarchical model in the statistical sense. Rather, it is a structured substitution model in which word segments are partitioned into subsets based on their phonological role, such as vowels of different types and consonant classes.

In the current implementation, the segment inventory is divided into ten such subsets. The model assigns one rate parameter for substitutions within each subset, and a separate rate parameter for substitutions between each pair of subsets, resulting in ten within-class rates and ten-choose-two between-class rates.

This structure allows the model to capture broad phonological regularities while remaining computationally tractable. We will revise the manuscript to make this construction more explicit and to clarify how the parameters of M3 are defined.

6) Reviewer 1

Comment

Concerning the environment influence. In Bouchard-Côté they use transducer to retranscribe some of these interactions. In your model another 1 possibility would be to use a different segmentation: for example align /k/ with /sj/, which would I guess be the best option here.

Response

We agree with the reviewer that a one-to-many correspondence such as /k/ to /sj/ is certainly relevant to sound change. As we explain in Section 5.4 of the manuscript, however, our current model lacks this capability, but we are planning to address this challenge in future work.

7) Reviewer 1

Comment

Another frequent assumption is the regularity of the changes (as you mention in 5.2, although the context cannot explain itself the regularity in a probabilistic point of view). As it is a common assumption. More concretely in the computation of the likelihood, without this assumption the proximity of two languages for which systematically /a/ corresponds to /e/ would be further away. Is the size of the dataset related with the arguably good results of the method? Or is it only because you select from three highly likely phylogenies.

Response

We interpret the reviewer's comment/concern as, "How does your probabilistic model encode the linguistic assumption of regular sound change? Does that regularity emerge naturally through shared parameters and repeated evidence, or are the good results mainly due to large data size or conditioning on already-likely trees?"

We thank the reviewer for raising this point, which touches on the notion of regular sound change and how it is represented in a probabilistic framework.

In our model, regularity is not imposed as an explicit constraint but emerges naturally through

parameter sharing in the likelihood. Segmental substitutions are governed by a single rate matrix that applies across all words, all alignment columns, and all branches of the tree. As a consequence, systematic correspondences—such as a recurrent /a/ to /e/ change between two lineages—are not treated as independent coincidences. Instead, repeated evidence for the same correspondence increases the posterior support for the corresponding substitution rate, which in turn increases the likelihood of trees that place those languages closer together. In this sense, regularity arises statistically through the accumulation of shared signal across many words.

If such systematic correspondences were absent, or if changes were genuinely irregular, the model would be unable to explain them parsimoniously with a small number of rate parameters, and the inferred distance between the languages would indeed be larger. Thus, the likelihood calculation directly rewards regular, repeated patterns of change, rather than assuming them a priori.

Regarding the role of dataset size, we agree that having multiple cognate sets is essential for identifying regularity in a probabilistic model. This is not a weakness specific to our approach but a general feature of statistical inference: regularity can only be inferred when sufficient repeated evidence is available. With very small datasets, any method—probabilistic or otherwise—will have limited power to distinguish systematic change from noise.

Finally, our results are not driven by the restriction to three highly likely phylogenies. Rather, conditioning on a small set of plausible trees was a pragmatic choice to make the computational problem tractable, given the problems we encountered with the MCMC procedure jointly exploring alignment and tree space. Within this set, the likelihood still discriminates strongly among models based on how well they explain the observed segmental correspondences. The superior performance of the Natural Class model reflects its ability to capture systematic patterns of change in the data, not the prior selection of tree topologies.

We added the following sentence to section 5.2 of the manuscript to clarify this point:

In this framework, regular sound change is not enforced a priori but arises statistically, since repeated segmental correspondences across many words increase the posterior support for the corresponding substitution rates and, consequently, for trees that place those languages closer together.

8) Reviewer 1

Comment

Another model that would be interesting to compare are the “phylogenies” inferred using Levenshtein distances, as your model corresponds to a weighted levenshtein distance by some aspects.

Response

We thank the reviewer for the suggestion. While our model involves insertion, deletion, and substitution processes at the segmental level, it is quite different from Levenshtein-distance-based approaches.

Levenshtein distance is a pairwise measure based on the single minimum-cost sequence of edits required to transform one string into another. By contrast, our framework defines a probabilistic model of sequence evolution along a phylogenetic tree, in which the likelihood

integrates over all possible alignments, ancestral sequences, and histories of insertion, deletion, and substitution, weighting each by its probability under the model. As a result, our method is not equivalent to a weighted edit distance, nor does it reduce to a distance-based inference procedure.

Moreover, Levenshtein-distance approaches require additional ad hoc choices, such as how to combine the distances in a tree, how to normalize the distances for word length, and how to handle uncertainty in homology. Our goal is not to approximate string similarity, but to perform likelihood-based inference under an explicit event-based model of word evolution.

For these reasons, we view Levenshtein-based phylogenies as addressing a different class of problems and not as a directly comparable alternative to the approach presented in this paper.

We added the following sentence to clarify this point to the reader.

Despite its use of insertion, deletion, and substitution processes, the model is not equivalent to a weighted edit distance, as it marginalizes over all possible evolutionary histories rather than optimizing a single edit sequence.

9) Reviewer 1

Comment

Assuming three topologies can be justified in the Romance example, but is not doable in general. Are there possibilities to infer fully the topology of the tree?

Response

We agree with the reviewer that conditioning on a small set of topologies is justified for the Romance example, but is not a general solution. Our choice to restrict attention to three plausible Romance topologies was motivated by computational considerations rather than conceptual ones.

In principle, the framework we introduce does allow for full inference of tree topology. In practice, however, we have found that joint inference of tree topology and segmental alignments under the TKF91 model is extremely challenging. In our experiments, Markov chain Monte Carlo chains reliably converged to high-likelihood trees, but then exhibited very poor mixing over tree space. Independent runs would become trapped on different, but similarly good, topologies. This behavior reflects a strong coupling between tree topology and alignment. Once a tree is fixed, the alignments rapidly adapt to it, creating high barriers in the probabilities that make moves to alternative trees unlikely.

We have explored several strategies to address this issue. One approach is to propose joint moves that modify both the tree topology and the alignments simultaneously, with the goal of reducing this coupling. While conceptually appealing, our initial experiments suggest that designing such proposals with reasonable acceptance rates is nontrivial, and this remains an area of ongoing investigation. A second, not yet implemented, possibility is to use multiple chains associated with different tree topologies, together with a higher-level MCMC variable that switches between them. In this scheme, the relative time spent in each topology would approximate its posterior probability. Such an approach would be substantially more computationally expensive and would require restricting the explored tree space to a manageable subset.

In short, while full topology inference is conceptually possible within our framework, making it

practical under a segment-level indel model is an open computational problem. We view the present work as a proof of principle, demonstrating that likelihood-based inference under a fully historical word-evolution model is feasible. Improving tree-space exploration under this model is an active area of our ongoing research.

We added the following sentence to clarify this point to the reader.

Although the framework in principle supports full Bayesian inference of tree topology, in practice the strong coupling between topology and segmental alignment under the TKF91 model makes efficient exploration of tree space computationally challenging, and improving this aspect of the method remains an active area of ongoing work.

10) Reviewer 1

Comment

A last question would be the set of possible sounds in the changes. What happens if the ancestral language contains sounds that have never been preserved but explains the strange patterns observed in the data?

Response

The reviewer raises the interesting issue of unobserved or extinct segmental states in ancestral languages.

In our framework, ancestral segmental states are not restricted to those observed in the modern languages. When computing the likelihood, the model marginalizes over all possible ancestral states and evolutionary histories consistent with the specified state space. Consequently, a segment may appear with nonzero posterior probability at internal nodes of the tree even if it is never preserved in any extant language, for example because it is systematically deleted or transformed along all descendant branches.

That said, the flexibility of the model is constrained by the choice of state space, which must be specified in advance. The model cannot, in its current form, infer entirely novel sound categories that lie outside this predefined inventory. However, one could in principle expand the state space to include hypothesized ancestral segments and allow different substitution and deletion processes to apply to different parts of the tree. Such nonhomogeneous, or lineage-specific, models are well established in molecular phylogenetics and provide a natural way to accommodate ancestral states that are present only in early lineages and later lost.

Exploring such extensions in a linguistic setting would be an interesting direction for future work, but beyond the scope of the present study.

11) Reviewer 1

Comment

Lexical innovation is particularly important in linguistics, in the example presented there is little to no innovation. Is that why the author chose this example? In Bouchard-Côté they infer jointly the cognacy classes and the transducers, would that be possible (or at least added as another step) in your model, or in future studies? There is an extensive literature on those models (Dollo model by Nicholls and Gray; Gray and Atkinson, etc.). I don't think these models are related directly to your article, but the second paragraph of 5.5 tells a different story.

Response [View changes](#)

In the papers the reviewer mentions, lexical innovation refers to a specific form of change, whereby a lexeme becomes the standard or neutral term for a concept. This type of lexical change is bound to a definition of cognacy in which semantics is relevant. (Under the Chang et al. typology, these are root-meaning traits and lexeme-meaning traits). One of the innovations of our approach is that we are no longer beholden to this definition of cognacy, which is in fact not the most prevalence definition of cognacy in historical linguistics. Historical linguists working in a qualitative framework focus on shared segmental ancestry when defining cognacy, with semantics playing little to no role. Our model is situated within this approach to cognacy and at present it is not clear whether it would bring any advantage to incorporate the cognate classes characteristic of root-meaning traits. What would be more in line with the spirit of our approach would be to combine phylogenetic inference with automatic cognate detection (where, again, cognacy is defined solely as shared segmental ancestry). This additional capability is one that we hope to add in future work.

12) Reviewer 1**Comment**

I don't understand this sentence in the conclusion "When applied to abstract cognate relationships, it is not clear what exactly is being counted along the branches." as the models I mention above are clear on what they count, the lack of clarity might be on the definition of the cognacy classes and their production from the original data. Note also that these models are event based as well.

Response

It is correct that the stochastic models typically applied in historical linguistics are event-based. In practice, these are k-state models, where k is determined during the prior step of cognate coding. For example, Figure 11.1 of Greenhill et al. (2020) outlines the Comparative Method as beginning with the assignment of discrete cognate states to languages. In their illustration, a single cognate set is coded with k equals 3 abstract states, which would then be analyzed using a three-state continuous-time Markov model, as implemented in tools such as MrBayes or BEAST.

While such models are formally event-based, the point raised in our conclusion concerns the interpretation of those events. The states themselves are abstractions produced by the coding process, rather than empirically observable linguistic forms. Consequently, when a transition occurs along a branch, for example from state B to state C, it is unclear what concrete linguistic change this event corresponds to. Does it represent a specific phonological or lexical transformation, or does it stand for a change involving forms that are not directly attested in the sampled languages? More generally, what linguistic entities are being gained, lost, or transformed by such a transition?

This is the sense in which we wrote that when applied to abstract cognate relationships, it is not clear what exactly is being counted along the branches. Our comment is not intended to deny that these models are event-based, but rather to highlight that the nature of the events depends critically on how cognacy classes are defined and abstracted from the original data. In contrast to models that explicitly specify the linguistic units undergoing change, the connection between model transitions and underlying linguistic processes is indirect, and in some cases, ambiguous.

13) Reviewer 1

Comment

The sentence “who also treat cognate word-forms as observations” concerning Bouchard-Côté, is false as they extend their method to the case of unknown cognacy classes as well. The sentence “model is a historical model with well-defined events that can transform a word” is also somewhat misleading, as the “event-based” model corresponds to a continuous time overall (and reciprocally on a probabilistic perspective).

Response

We believe the reviewer may be conflating two different modeling distinctions, which is the source of the apparent disagreement.

First, regarding Bouchard-Côté et al. (2013), we agree that their method allows cognacy classes to be unknown a priori and inferred during analysis. However, this does not change the nature of the observations in their likelihood. In their framework, the observed data are still the surface word forms, or segments thereof, at the tips. Cognacy is treated as a latent structure linking these observed strings, not as an additional type of observation. In this sense, our statement that they treat cognate word-forms as observations was meant to contrast surface linguistic forms with abstract cognate states, not to deny that cognacy itself can be inferred.

Second, we respectfully disagree that their transducer-based model is a historical model in the same sense as models such as TKF91. The string transducer used by Bouchard-Côté et al. defines a probability distribution over pairs of strings via weighted edit operations, but it does not explicitly represent a sequence of evolutionary events unfolding along branches of a phylogenetic tree. That is, it assigns probabilities directly to string transformations, rather than integrating over explicit insertion, deletion, and substitution histories occurring through time. In this sense, it is not a historical model, even though it is motivated by diachronic considerations.

By contrast, the TKF91 model is explicitly historical. When computing the likelihood of the observed word forms at the tips of a tree, the model integrates over all possible sequences of insertion and deletion events along each branch, governed by a continuous-time birth–death process. The events are not merely a probabilistic abstraction but correspond to well-defined operations acting on evolving sequences through time.

Regarding the reviewer’s second point, we agree that event-based models such as TKF91 operate in continuous time and that events are not observed individually. However, this does not undermine our claim that the model has well-defined events. Rather, it is precisely the explicit specification of those events and their rates that distinguishes historical sequence-evolution models from abstract state-based or transducer-based approaches. Our intent was to emphasize this distinction, not to suggest that events are discretely observed or temporally localized in the data.

14) Reviewer 1

Comment

Overall, I wonder if the example of romance languages is adapted: complexity of the latin forms, addition of these empty characters that depends on the choice of the prior on H, interaction of sounds, multiplicity of the forms for a single meaning (e.g. accusative/nominative, masculine/feminine), change of the meanings. In the results you observe that the deletion and insertion are more frequent than substitutions. Doesn’t that mean that the model just favors insertion and deletion: if there are only half as much transitions, wouldn’t that just be a problem of prior? The other possibility is that there are lots of insertion and deletion in your dataset overall. Maybe the question should be the segmentation and the alignment themselves.

Response

For further clarification of the segmentation and alignment, see issues 1 above and 18 and 19 below. The greater frequency of the insertion and deletion rates is driven by the empirical frequency of these events in the data.

15) Reviewer 1**Comment**

In Fig 11 why is the alignment different depending on the models? Is that because it is easier to align in some way if the rates are different between the classes (in particular deleting a /k/?).

Response

In our framework, the alignment itself is part of the inference. Each alignment column represents a hypothesis about homology, that is, which word segments in different languages descend from a single ancestral segment. We therefore treat the alignment as a random variable and use Markov chain Monte Carlo to sample alignments in proportion to their posterior probability under a given evolutionary model.

It is therefore both expected and informative that the posterior distribution of alignments differs across models. The reason is that the probability of any proposed alignment depends directly on the substitution and indel processes assumed by the model. Different models assign different probabilities to the same homology hypotheses.

To make this concrete, consider an alignment column in which two segment states, say 0 and 1, are observed across languages. Such a column necessarily implies at least one historical change along the tree, either a direct substitution between 0 and 1 or a sequence of changes through intermediate states. The probability of that column is therefore sensitive to the rate matrix of the continuous-time Markov model governing substitutions. If transitions between the relevant states, or via intermediates, are relatively frequent, the alignment column will be more probable. If they require rare changes, it will be less probable.

The same logic applies to deletions. If a model assigns a relatively high deletion rate to a particular segment, such as /k/, alignments in which that segment is absent in some languages become more probable under that model. Conversely, models in which such deletions are rare will favor different alignment structures. Thus, alignments that appear easier under one model are simply those that are more probable given that model's assumptions about rates of substitution and deletion.

In Figure 11, the differences in inferred alignments therefore reflect differences in the underlying rate assumptions. The simplest model, isomorphic to JC69, assigns equal rates to all changes, whereas the more complex models allow rates to vary across segment types and transitions. These assumptions propagate naturally into the posterior over alignments, leading to the observed differences.

In short, the alignment differs between models precisely because the alignment is being inferred jointly with the evolutionary process, and different models encode different notions of what kinds of historical changes are likely.

16) Reviewer 1

Comment

I would have liked to see some examples on simulated dataset, or even on another real family of languages (which is much more work so I understand to only focus on a single family).

Response

We address the issue of a simulation study in C30, below.

17) Reviewer 1

Comment

P6 - Finally, the equilibrium frequencies reveal that Romance languages rely primarily on stops and short vowels for phonemic contrasts" I don't understand how this relates with the previous sentence: these equilibrium frequencies mean that the stops are more frequent, not that they contrast more, or you need additional steps of thinking. I don't really know why those frequencies are interesting.

Response

We thank the reviewer for this comment, which made us realize that the original formulation came to close to conflating frequencies and functional load. We have adjusted the text to make it clear that our results do not bear on the question of functional load.

18) Reviewer 1

Comment

I think the theoretical part on Bayesian is very accessible to the non-statisticians, which is a good point. I think more details should be added on the homology H: what is the prior chosen on it? What is the exact procedure you use to compute its posterior distribution (I assume you integrate along the MCMC sample the posterior given the parameters, but it would be good to write it).

Response

The homology structure H is treated as a latent variable and is sampled jointly with all other model parameters using Markov chain Monte Carlo. Its posterior distribution is therefore obtained from the MCMC output rather than by post hoc integration over samples of the remaining parameters.

As described in the Appendix, the prior on H is induced by the TKF91 insertion–deletion process and corresponds to a geometric distribution on alignment length with parameter λ divided by μ , which is the equilibrium length distribution of the model. This prior assigns higher probability to compact alignments when insertion rates are low relative to deletion rates.

To update H during MCMC, we use the proposal mechanism described by Lunter et al. (2004). At each update, a random contiguous block of alignment columns is selected, removed, and resampled under the TKF91 model conditional on the surrounding context.

The proposed alignment for this region is then accepted or rejected according to the Metropolis–Hastings criterion, using the ratio of posterior probabilities of the proposed and current alignments. This procedure defines a valid Markov chain whose stationary distribution is the joint posterior over tree, alignment, and model parameters.

In this way, the posterior distribution of H is obtained directly from the MCMC sample and can be summarized by any desired posterior statistic, such as marginal alignment probabilities or posterior homology support between segment pairs.

19) Reviewer 1

Comment

The more I think about it the more I think the prior on H should depend on the value of λ and μ . But in this case, why is it a marginal prior and not a prior given these parameters? (p7 appendix) Please add some detail on that, as it might also explain why you observe more deletions and additions.

Response

The reviewer is absolutely right that, from a generative perspective, the prior on the homology structure H should depend on the insertion and deletion rates λ and μ . Under an explicit indel process, these parameters determine the expected sequence length and, consequently, the prior distribution over alignment lengths and homology patterns. In particular, when insertion rates are low relative to deletion rates, shorter and more compact alignments should indeed receive higher prior probability than very long alignments.

Upon re-examining our implementation of the TKF91 model, we realized that our prior on H is not uniform, as implied by the manuscript text. Instead, the prior probability of an alignment length follows a geometric distribution with parameter λ divided by μ , corresponding to the equilibrium length distribution of the TKF91 model. Thus, alignments with many columns are already down-weighted a priori, and the prior on H does depend on λ and μ in an important and meaningful way. In this sense, the current implementation is closer to a generative indel-based prior than our description suggested.

That said, the prior is still not fully generative in the strongest sense, because it does not explicitly model the full homology structure H as a realization of a TKF91 process conditional on λ and μ . Rather, it captures the equilibrium length distribution implied by the model, while remaining agnostic about finer-grained dependencies among homology patterns. We agree that a prior constructed by explicitly generating H under the TKF91 process would be more coherent. However, implementing such a prior in a computationally tractable way, particularly in the context of joint inference over trees, alignments, and substitution parameters, remains challenging.

We will revise the manuscript to correct the description of the prior on H and to clarify that alignment lengths are geometrically distributed with parameter λ divided by μ , as implied by the equilibrium behavior of the TKF91 model. We will also clarify that developing fully generative priors on homology structures conditional on indel parameters is an important direction for future work.

We have added the following sentence to the manuscript.

A more coherent alternative would be to define a fully generative prior on H conditional on λ and μ , building directly on the TKF91 process itself. Developing such priors for segmental homology structures remains an important direction for future work.

We also added a sentence clarifying the prior on the alignment:

An alignment of length L receives a prior probability that is geometric with probability λ / μ . (Under the TKF91 model, at equilibrium, the length of an alignment is geometrically distributed with parameter λ/μ .)

20) Reviewer 1

Comment

The article by Lunter is quite difficult to read as it includes several elements linked with genetics, and I think some more details in supplement would be great.

Response

We agree with the reviewer that the paper by Lunter et al. is dense and difficult to read. This is largely due to the mathematical and algorithmic complexity of the problem it addresses rather than its connection to genetics per se.

The central contribution of Lunter et al. is not tied to molecular biology, but rather to the development of an algorithm for computing the likelihood under the TKF91 model on an arbitrary phylogenetic tree. Prior to that work, likelihood calculations under TKF91 were only available for pairs of sequences.

The fact that Lunter et al. apply their method to DNA sequences is incidental. The TKF91 model consists of a continuous-time birth–death process governing insertions and deletions, coupled with an arbitrary continuous-time Markov substitution process on surviving characters. Any substitution model can be used in place of DNA evolution, including the phonological substitution models used in the present study.

For this reason, we view the genetics context of the Lunter et al. paper as largely orthogonal to its algorithmic contribution. What is essential for our purposes is the dynamic programming method it provides for marginalizing over all possible insertion, deletion, and substitution histories on a tree.

While we agree that additional exposition of this algorithm could in principle be helpful, we are concerned that reproducing its full derivation in the supplement would substantially increase the length and technical density of the paper without improving accessibility for most readers. Readers with a strong interest in the algorithmic details are encouraged to consult the original paper and the accompanying software implementation.

We also note that although the TKF91 model is historically important as the first fully event-based indel model, it has not seen widespread adoption in molecular evolution, largely because biological insertions and deletions typically involve multi-nucleotide events rather than single-site changes. Nevertheless, TKF91 remains a foundational model in the literature and provides a mathematically clean framework for studying sequence evolution under insertion and deletion.

For linguistic data, where segmental insertions and deletions are more naturally modeled at the single-segment level, the TKF91 framework is particularly well suited and provides a principled starting point for historical modeling.

21) Reviewer 1

Comment

The sentence “the likelihood accounts for all the histories of insertion, deletion, and segmental substitution that could have led to the observed words sampled from the nine languages” in 5.1 is strange to that regards. If it accounted for all the additions and deletion possible, I think the number of possible H 3 would be infinite, as the ancestral state can have any number of deleted characters. There is an approximation here somewhere (which is fine but would be interesting to understand exactly).

Response

The reviewer is correct that, if considered naively, the space of possible insertion–deletion histories is infinite. An ancestral sequence could in principle contain arbitrarily many characters that are later deleted. However, this does not invalidate our statement. The likelihood we compute does indeed account for all possible histories of insertion, deletion, and substitution, weighting each by its probability under the model. This is not an approximation, but a standard consequence of marginalization in continuous-time stochastic processes.

Formally, the likelihood marginalizes over all unobserved evolutionary histories. Although the set of such histories is infinite, their total contribution is finite because histories with many insertion and deletion events have vanishingly small probability under the model. This is entirely analogous to other continuous-time Markov processes, where infinitely many event sequences are possible over any positive time interval, yet transition probabilities remain well defined.

This is the same principle that underlies likelihood calculations in simpler substitution-only models. In programs such as MrBayes or BEAST, the likelihood integrates over all possible assignments of ancestral states at internal nodes via the Felsenstein pruning algorithm, and all possible numbers and timings of substitution events along each branch via matrix exponentiation. Even in these models, the likelihood implicitly accounts for infinitely many substitution histories, despite the fact that only a finite state space is observed.

For models with insertions and deletions, such as TKF91, the situation is conceptually identical but computationally more challenging. The TKF91 process defines a continuous-time birth–death model on sequence length, coupled with substitution on surviving characters. Although an ancestral sequence may be arbitrarily long, the equilibrium length distribution is proper, and the probability of extremely long ancestral sequences decays rapidly. The dynamic programming algorithm of Lunter et al. (2003) exploits this structure to marginalize analytically over all possible indel histories without enumerating them explicitly. The poor scaling of the algorithm reflects the difficulty of performing this marginalization exactly, not the presence of an approximation.

Thus, while the reviewer’s intuition that there must be infinitely many histories is correct, this does not imply that the likelihood is ill-defined or approximate. On the contrary, the likelihood is precisely defined as a sum or integral over all such histories, and this is exactly what our statement in Section 5.1 was meant to convey.

22) Reviewer 1

Comment

I think the paper is interesting, but I have the impression that most of the choices are ad hoc: inserting the empty

characters before the /a/ so that it can evolve into /ie/ in french really does not seem possible unless the people using the model have a thorough knowledge of the history of the languages. It is my believe that comptuational methods in linguistics should be used to reduce human workload, and to get quantitative results. On the second aspect the method proposed here is interesting, but I seriously doubt the possibility to apply it in less studied languages, where adding those empty characters would be critical.

Response

There are no empty characters in the representation of cognate word-forms, in the sense that the absence of a character is not a possible character state. The dashes in the datasets are actual gaps. As we explain in the paper, the model requires an initial input of aligned phonemic representations, some of which contain gaps as way to represent, e.g., the correspondence between /a/ and /ie/. These are only initial inputs, however. As described in the manuscript, the model marginalizes over all possible segmental alignments, so that none of the parameter estimates assume any particular segmental alignment. In other words, one does not actually have to know that /a/ evolves into /ie/. One only has to know that the lexemes themselves are cognate. One does not need to know the homologies of the individual segments. Our approach therefor does reduce human workload, since it spares researchers from having to make decisions about segment-level homologies. In addition, there is no barrier to using our method with less-studied languages, provided that lexeme-level cognacy is known.

23) Reviewer 1

Comment

p7: lieasinabed, the whole list should go in appendix.

Response [View changes](#)

Moved list to the end of the Supplemental section.

24) Reviewer 1

Comment

Conclusion: estimation

Response [View changes](#)

Corrected.

25) Reviewer 1

Comment

I don't think Ψ is an amazing notation for a tree.

Response

We appreciate the reviewer's comment regarding notation. Our use of the symbol Ψ to denote a tree is not intended to be aesthetic, but conventional. In the phylogenetics literature, it is common to represent a phylogenetic tree as the combination of a topology, typically denoted by tau, and a set of branch lengths, often denoted by v. The pair is then written compactly as $\Psi = (t, v)$. This notation is used, for example, to distinguish between the discrete combinatorial structure of the tree and its continuous branch-length parameters.

We adopted this convention to make the distinction between topology and branch lengths

explicit and to remain consistent with existing usage in Bayesian phylogenetic modeling. That said, we are not wedded to this choice of symbol and would be happy to change the notation if the editor or reviewer strongly prefers an alternative.

26) Reviewer 1

Comment

Supplement p6: I would suggest introduce notations X for the observations and θ for the parameters. That would make everything more readable. Note that in your example, the tree topologies are not the parameters per se, they are the possible values of the parameter τ (same for ν). The assumption $\lambda < \mu$ might be better as part of the prior distribution.

Response

In the manuscript we use S to denote the observed sequences of segments sampled for a cognate across the Romance languages, and we reserve Greek letters for model parameters. This convention is consistent with standard usage in phylogenetics and allows us to distinguish clearly between observed linguistic data and latent model parameters.

We agree that, formally, the tree topology and branch lengths are values taken by parameters τ and ν , respectively, rather than parameters themselves. However, for readability we often refer to a particular realization of these parameters simply as “the tree,” which is conventional in the phylogenetic literature.

With regard to the suggestion to introduce X for the observations and θ for the parameters, we prefer to retain our current notation, as it makes explicit the linguistic nature of the data and keeps the connection to the evolutionary model transparent. We do not believe that introducing additional abstract symbols would substantially improve clarity for the intended readership.

Finally, the constraint $\lambda < \mu$ is already enforced naturally by our prior specification. Both λ and μ are assigned independent and identical exponential priors, with support restricted to the region $\lambda < \mu$, so that the prior density is zero for parameter combinations with $\lambda \geq \mu$.

27) Reviewer 1

Comment

If you want a more accurate name for Metropolis-Hastings, you can use the full name Hastings-Metropolis-Rosenbluth-Teller (which has the advantage of giving back credit to the original authors).

Response

We appreciate the reviewer’s comment and agree that proper attribution is important. The original 1953 paper indeed involved multiple contributors. Metropolis introduced the core idea of simulating a physical system via a Markov process with a desired stationary distribution. The Rosenbluths and Teller contributed substantially to the mathematical formulation and practical implementation of the method. Hastings later generalized the original algorithm in a probabilistic framework, leading to what is now widely known as the Metropolis–Hastings algorithm.

That said, while the longer name Hastings–Metropolis–Rosenbluth–Teller more fully reflects the authorship of the original work, the term Metropolis–Hastings is overwhelmingly standard in the statistics, physics, and computational biology literature. Using a nonstandard name would likely cause confusion for readers rather than clarify attribution.

For these reasons, we have retained the conventional terminology, while fully acknowledging the broader historical contributions underlying the method.

28) Reviewer 1

Comment

Appendix 1: You should precise “size of the 95% set ...” why are some of the meanings given in english and other in english and latin?

Response

We have removed the Latin meanings.

29) Reviewer 1

Comment

Last paragraph of 5.1, avoid naming a single number with several letters GM(L) could be written with a single letter. 4

Response

The symbol was changed to $\bar{L}$

30) Reviewer 2

Comment

The authors introduce a framework for applying the TKF91 model of molecular evolution of DNA sequences for comparative linguistic analysis at the segment and word levels. Using Bayesian phylogenetic approaches, inference is drawn from cognate word-forms and their segmental compositions directly. They use a subset of Romance languages as a test case, compare three different models (Poisson, an isomorphic DNA sequence evolution, Natural Class model), and results show that the Natural Class model explains the data much better than the other two models.

The introduction does a nice job of laying out the problems with current historical-comparative approaches, e.g., "the transformation of word-forms into abstract cognate relationships results in pervasive information loss", "since root-meaning traits are analyzed the same way biologists analyze morphological characters ... they are subject to the same limitations", "transformation of multistate cognate relationships into binary characters introduces dependencies into the data", "continuous-time Markov chains that model the evolution of lexical items in cognate classes assume that character evolution is independent and identically distributed".

For segmental changes, they find a clear division between vowels and consonants, as expected -- with vowels being more subject to change and diphthongs and long vowels the least stable. This coincides with what I know in the literature.

Response

We thank the reviewer for this clear and accurate summary of our work and for the positive assessment of the motivation and results. We are particularly pleased that the reviewer finds the framing of the limitations of existing comparative approaches convincing, and that the empirical findings, especially the distinction between vowels and consonants and the relative stability of diphthongs and long vowels, are consistent with established results in the linguistic literature.

With respect to a simulation study, we fully agree that simulation-based validation, such as Simulation-Based Calibration, is a powerful and principled way to assess the behavior of Bayesian inference procedures, particularly for novel models. In an ideal setting, such a study would indeed be desirable.

In the present case, however, a comprehensive simulation study is not practically feasible. Each full Markov chain Monte Carlo analysis under the TKF91-based framework requires multiple

days of computation on a modern desktop computer. A properly designed simulation-based calibration analysis would require on the order of thousands to tens of thousands of independent simulations, each followed by a full MCMC run to convergence. At current computational costs, this would translate into months or even years of compute time, well beyond what is realistic for this study.

That said, the absence of a large-scale simulation study does not imply a lack of theoretical grounding. The inference framework we employ is standard Bayesian likelihood-based inference, applied to a new class of data rather than a new inferential paradigm. Under the assumed model, Bayesian posterior distributions have well-understood and desirable theoretical properties, including correct uncertainty quantification and asymptotic calibration. In particular, when the model is correctly specified and the posterior is adequately explored, Bayesian credible intervals contain the true parameter values with their nominal probabilities in repeated sampling.

Moreover, the likelihood calculations themselves are exact under the TKF91 model, in the sense that they marginalize analytically over all unobserved histories of insertion, deletion, and substitution, rather than relying on approximations or heuristic scoring schemes. The computational burden arises precisely because of this exactness. The tradeoff between computational tractability and model realism is a familiar one in phylogenetics, and our work deliberately prioritizes model fidelity over speed.

We view the present study as a proof of principle demonstrating that fully historical, segment-level evolutionary models can be applied directly to linguistic data within a Bayesian phylogenetic framework. We agree that future work, benefiting from algorithmic improvements, parallelization, or approximate inference schemes, should include extensive simulation-based validation, and we see this as a natural and important direction for further research.

31) Reviewer 2

Comment

I was a bit confused by the remarks regarding primal consonants:

- "There are two important differences between their primal consonants and our results, however.
- First, these consonants are deemed primal because they are only rarely created by sound change.
- In other words, they are rarely the output of sound change.
- Under our framework, by contrast, segmental stability is associated with the input of sound change in that the diagonal of the Q matrix reflects the rate of transition away from a given phoneme."

It could also be argued that they are only rarely created by sound change because most languages have these consonants. If they are typologically common, then they are necessarily rare outcomes of sound change. When they are missing, e.g., bilabial plosives, they are typically very quick to be reintroduced into a phoneme inventory. In this regard, the authors might want to pay a bit of lip service to the cross-linguistic typological frequency of segments and its effects on borrowing and sound change in regards to their model, e.g.:

- <https://www.science.org/doi/pdf/10.1126/sciadv.adv7521>
- <https://dx.doi.org/10.1353/lan.2024.a947038>
- <https://doi.org/10.1371/journal.pone.0242709>

But that's not necessary in my opinion for publication, but might be nice in light of comments such as: "While our results are relevant for the typology of diachronic phonology in that they capture broad trends in segmental evolution, they should not be interpreted as universal patterns of sound change." I think they do a nice job of outlining the limitations of the current model, without getting into too much detail, e.g.: "current implementation of the model is unable to take into account phonetic variation of word-forms" and in describing the limitations of what the model can do in regard to sound change patterns that aren't one-to-one (in Section 5.4).

Response

We thank the reviewer for these comments and for our attention to these articles. We have clarified the description of primal consonants, but have decided not to discuss the literature on borrowing and sound change in this study.

32) Reviewer 2**Comment**

I find the referencing of figures a bit jarring, i.e., they aren't necessarily in order (e.g., 1, 2, 3...)

Response

The figures have been redistributed more evenly throughout the length of the manuscript.

33) Reviewer 2**Comment**

Colors in figure 1 aren't color blind friendly (I also find the two greens hard to distinguish)

Response

Changed ambiguous colors to color-blind friendly ones.

34) Reviewer 2**Comment**

"(DÉRom, 2023)"?

Response

N.B. the ATILF is getting rid of the Stella-server, so this URL will soon no longer work, whereas the "generic" URL will always be maintained.

35) Reviewer 2**Comment**

"study of (Moran et al. 2021, pp. 92-95)"

Response

Corrected

36) Reviewer 2**Comment**

- "they are measuring the rate at which the size of segmental inventories changes"

Response

Changed the phrasing.

37) Reviewer 2**Comment**

I just have a hard time parsing this: "on the basis of which three robust properties of sound change"

Response

This sentence has been corrected in the manuscript.

38) Reviewer 2**Comment**

"not ASCII or ANSI" -- why is ANSI mentioned here (or at all)?

Response [View changes](#)

That was a good catch. We said 'ANSI' when we meant the 'ANSI Latin 1' code page. We rephrased this part to make it more precise.

39) Reviewer 3**Comment**

Given the novelty of the TKF91 model to linguistic data, I was surprised that no simulation study was provided. The model should be validated using Simulation-Based Calibration.

Response

See our response to Comment 30.

40) Reviewer 3**Comment**

The best-fitting model M3 is heavily parameterized relative to the dataset size (133 parameters for the substitution model estimated from only 102 cognate sets in 9 languages). I wonder if this is a case of overfitting to the particular historical trajectory of Romance. For example, the low equilibrium frequency estimated for diphthongs may simply mirror their monophthongization from Latin in this sample rather than reflect a general property of diphthongs in language change. The rate differences in Figure 7 are quite extreme, which made me think that perhaps very low equilibrium frequencies are being assigned to phonemes that just happen to be rare. If datasets of the same size as the empirical data were simulated under a model with less extreme values of stable frequencies and exchangeability rates, would some phonemes be rare simply due to chance? Would the analysis estimate a low equilibrium frequency for these phonemes, even when the true generating frequency is higher?

Response

Overfitting is unlikely to be an issue since the natural class model contains only a fraction of the possible number of parameters (i.e., one for each possible transition among the 79 unique segments). We have also made it more explicit that all of the results regarding sound change are specifically about Latin and Romance, and not sound change. The low equilibrium frequency for diphthongs is unquestionably a reflection of the monophthongization processes in Latin and Romance and not a general property of sound change.

41) Reviewer 3**Comment**

I like the argument that in a typical linguistic phylogenetic analysis, cognate classifications are used as observations even though they are hypotheses. However, I don't see how this new approach gets away from the problem. While the method analyzes observable segmental data, the dataset still depends on traditional cognacy judgments to decide which word-forms are grouped together. Any misclassification at this stage propagates into the analysis, just as in conventional approaches.

Response

We are grateful to the reviewer for this comment. We have tempered the original sentence to make it clear that the input data also depend on cognate judgments, even though they do not require transformation.

42) Reviewer 3**Comment**

In Figure 5, the clustering of -qii values by natural class is striking, but doesn't it just follow from the model structure? In the Natural Class model, all phonemes in a given class share the same exchangeability rates. Considering two phonemes in the same class, the rows of the Q matrix will be almost identical apart from the difference of 1 out of the 79 stable frequency values. In other words, figure 5 is driven by the exchangeabilities (shared within class) not the equilibrium frequencies. It seems to me the equilibrium frequencies would have to be drastically different within classes to produce a different pattern than that seen in figure 5. The authors should discuss the extent to which clustering is expected given the model structure.

Response [View changes](#)

We have clarified the relationship between the -qii values and the rate parameters. The clustering among rates is partially due to the model structure and partially to the similarity in equilibrium frequencies.

43) Reviewer 3**Comment**

The code repository lacks any instructions or a readme file. The repo should be updated with the aim that the model can be applied by other researchers to their own data.

Response

The repository now has a "readme.pdf" file.

44) Reviewer 3**Comment**

There appears to be a contradiction in the parameter count for the Natural Class model. Earlier, the text states that there are "79 parameters for the equilibrium distribution and 80 parameters describing rates among natural classes," whereas later the model is described as having 78 equilibrium parameters plus 55 class-rate parameters (total 133). The latter seems correct given the 10-class structure (45 off-diagonals + 10 within class rates), so is 80 a typo?

Response [View changes](#)

We apologize for this mistake, and have corrected the text. For the equilibrium parameter, there are 79 frequencies, one for each segment state in our model. But, there are only 78 free parameters, because the last frequency can be determined by the sum of the other 78 frequencies. (Remember, the frequencies are constrained to sum to one. Under the Natural Class model there are, indeed, 55 rate parameters, 10 for the changes within the classes and 45 parameters to describe the rates of change between the classes.

45) Reviewer 3**Comment**

The authors filtered out any cognate sets without a form in one or more languages. I am unsure after reading the paper

whether or not this was a data collection decision, or if the method implementation actually requires complete data like this. Please clarify.

Response [View changes](#)

We have clarified in the manuscript that incomplete cognate sets can be used but that we only used complete ones to simplify the modeling for this initial study.

46) Reviewer 3

Comment

The section starting around Figure 3, which presents an IPA–Unicode encoding example and raw data for the concept “mountain”, is informative but hard to follow in terms of the overall data-processing path. I recommend adding a schematic diagram of the workflow, from initial cognate identification through IPA encoding, segment parsing, alignment generation, and Bayesian inference to final rate and topology estimates. Such a figure would help readers understand the broader computational pipeline, and clarify where key modelling and data decisions occur.

Response

A 'Workflow' section was added to the Supplemental to describe the experimental processing. A pipeline figure was created and put in the same section.

47) Reviewer 3

Comment

The paragraph on page 3 beginning “Second, since root-meaning traits...” seems to be a criticism of the multistate model for cognate evolution. This is confusing for two reasons. First, it has already been stated on the previous page that the data is first binarised. Second, “state 0 of one cognate is not equivalent to state 0 of another”, surely you mean state 0 of a concept? The multistate approach has been attempted a few times (Heggarty et al. 2023 SI, Rönchen et al. 2024 and references therein), but for the most part linguistic phylogenies work on binarised data.

Response [View changes](#)

We have now made it clear in the manuscript that the criticisms in this paragraph apply to both binary and multistate encoding. We have also corrected the typo.

48) Reviewer 3

Comment

The authors state that the method works on cognate words regardless of meaning. However, the data is organised by meaning. I assume that this just refers to the meaning in Latin, but please clarify.

Response

We have clarified in the manuscript that the organization of the cognate sets by meaning is purely a way to help keep track of the data.

49) Reviewer 3

Comment

It may help the reader if Figure 7 is labelled explicitly as showing class-pooled average rates derived from the full 79×79 Q-matrix of phoneme-to-phoneme substitutions. Currently, the relationship between the full Q and the reduced 10×10 class matrix in the figure is not stated clearly. Same for Figure 6, I guess these are pooled from the individual phonemes.

Response

We revised the text to indicate that the figures show pooled rates.

50) Reviewer 3**Comment**

It may be worth defining “unigram” explicitly in the manuscript, as many readers from historical linguistics rather than computational linguistics may not be familiar with the term

Response

We have removed the term unigram here.